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Mechanism-aware model selection for food-quality deteriorationunder repeated measures: a Robustness-Integrated Evidence andChoice (RIEC) benchmark on olive oil UV indices

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Re: Submission of manuscript: "Mechanism-aware model selection for food-quality deterioration under repeated measures: a Robustness-Integrated Evidence and Choice (RIEC) benchmark on olive oil UV indices"

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Dear Editor,

We would like to submit the above manuscript for consideration in *Current Research in Food Science*. Food-quality deterioration datasets increasingly contain repeated measurements for each product unit (e.g., each oil), forming correlated within-unit trajectories and often exhibiting phase transitions. These properties complicate model selection: information criteria can miscalibrate complexity when observations are dependent, while predictive-only cross-validation can drift toward over-flexible candidates for tiny error gains.

In this work, we propose **Robustness-Integrated Evidence and Choice (RIEC)**, an audit-ready selection protocol that (i) corrects information content via an effective sample size under repeated measures, (ii) integrates baseline-normalized out-of-sample evidence (XPE) with a complexity penalty, and (iii) enforces decision-aligned evaluation regimes (forward-by-time forecasting and group-by-oil generalization), using random splits only as a negative control for trajectory leakage. We benchmark RIEC on a public olive-oil ageing dataset and model three UV indices (K232, K268 and ΔK) using a small, interpretable candidate family.

From a food-industry/QC perspective, the protocol helps avoid overconfident shelf-life modelling decisions arising from misaligned evaluation or overfitting, and provides a standardized reporting workflow that improves comparability across studies and laboratories. We do not claim a new chemical mechanism; rather, we provide a practical, mechanism-aware model-selection wrapper that aligns with common deterioration decision contexts.

The manuscript has not been published and is not under consideration elsewhere. All authors have approved the submitted version. The authors declare no competing interests. The dataset is publicly available (Mendeley Data DOI: 10.17632/g6y69g8gwm.1), and a runnable reproduction package is provided via GitHub (https://github.com/lijincheng494gmail/crfs-reproduce).

Sincerely,

Jincheng Li
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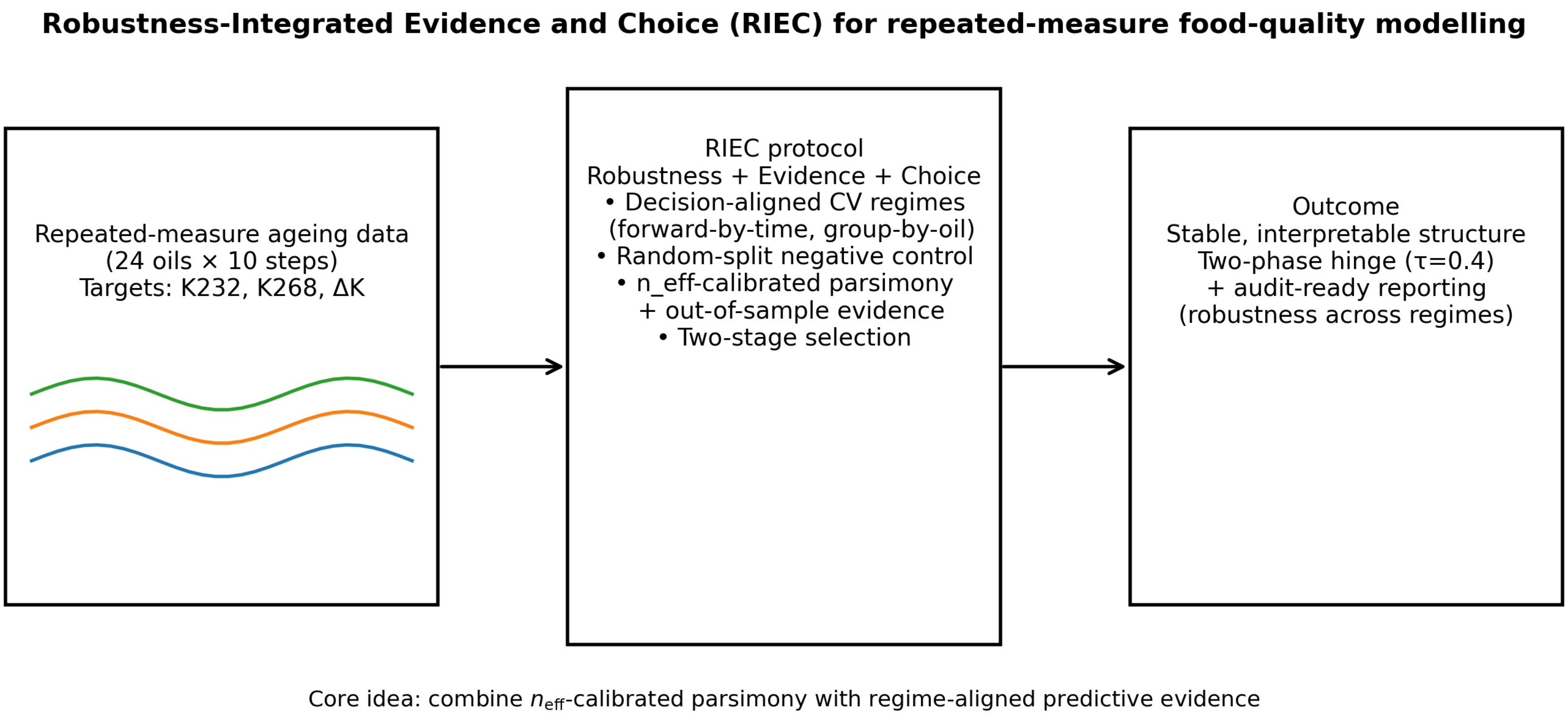
Robustness-Integrated Evidence and Choice (RIEC) for repeated-measure food-quality modelling
Repeated-measure ageing data
(24 oils × 10 steps)
Targets: K232, K268, ΔK
RIEC protocol
Robustness + Evidence + Choice
• Decision-aligned CV regimes
(forward-by-time, group-by-oil)
• Random-split negative control
• n_eff-calibrated parsimony
+ out-of-sample evidence
• Two-stage selection
Outcome
Stable, interpretable structure
Two-phase hinge (τ=0.4)
+ audit-ready reporting
(robustness across regimes)
Core idea: combine n_{eff}-calibrated parsimony with regime-aligned predictive evidence

Highlights

- Repeated-measure trajectories can mislead CV/BIC model choice in food deterioration.
- RIEC combines n_eff-calibrated parsimony with baseline-normalized CV evidence.
- Decision-aligned CV regimes plus random-split control diagnose trajectory leakage.
- On olive oil UV indices, RIEC selects a robust two-phase hinge structure (τ=0.4).
- CV-only can drift to extra flexibility for tiny gains under group-by-oil testing.

Mechanism-aware model selection for food-quality deterioration under repeated measures: a Robustness-Integrated Evidence and Choice (RIEC) benchmark on olive oil UV indices

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ABSTRACT

Food-quality deterioration datasets increasingly contain repeated measurements for each product unit, forming correlated within-unit trajectories and often exhibiting stage transitions. These properties complicate model selection: information criteria can miscalibrate complexity when observations are dependent, while cross-validation can prefer over-flexible candidates for small error gains. We present **Robustness-Integrated Evidence and Choice (RIEC)**, a mechanism-aware model selection protocol for finite candidate families. RIEC (i) evaluates candidates under decision-relevant cross-validation regimes and uses a random-split negative control to diagnose trajectory leakage (*Robustness*), (ii) integrates a BIC-style parsimony term based on an effective sample size with an out-of-sample evidence term derived from baseline-normalized CV error (*Evidence*), and (iii) performs an explicit two-stage selection that separates structural assumptions from within-structure complexity (*Choice*).

Using an open olive-oil ageing dataset (24 oils, 10 ageing steps; $n = 240$), we model UV absorbance indices (K232, K268 and ΔK) as oxidation proxies. We design three evaluation regimes to reflect distinct decision contexts: forward-by-time extrapolation, group-by-oil generalization, and a random split as a negative control. The measurement layer estimates strong within-oil dependence ($\rho = 0.500$) and an effective sample size ($n_{\text{eff}} = 43.6$), which materially changes complexity penalties under repeated measures. Under forward-by-time evaluation, RIEC selects a two-phase hinge model ($\tau = 0.4$) and achieves $E_{\text{CV}} = 0.0496$ with XPE $= 1.893$, corresponding to a 47.2% error reduction versus an intercept baseline. Structural choice is consistent under group-by-oil evaluation, whereas random splits yield optimistic errors and unstable rankings. RIEC provides an audit-ready workflow for repeated-measure deterioration modelling without claiming a new chemical mechanism.

1. Introduction

Accurate modeling of food-quality deterioration is central to quality control, process optimization, and shelf-life management. In lipid-rich products such as edible oils, oxidation-related indicators often evolve nonlinearly and may exhibit a lag phase followed by acceleration [7]. Public datasets increasingly provide repeated measurements over time for each product unit, forming within-unit trajectories rather than independent observations [4, 5]. When trajectories are treated as independent samples, both uncertainty quantification and complexity penalties can be miscalibrated. Model selection becomes particularly sensitive when researchers must choose between mechanistic structures and flexible empirical fits.

In food-industry settings, the cost of model misspecification is rarely summarized by a single average error metric. Quality control (QC) decisions often depend on predicted times to cross regulatory or internal acceptance thresholds (e.g., IOC UV limits for extra virgin olive oil) and on whether a model extrapolates conservatively under limited information [6]. Over-flexible candidates can absorb within-unit noise and shift the estimated "time-to-limit", which may increase the risk of shelf-life mislabelling, avoidable product holds/recalls, or non-compliance, while overly conservative choices can shorten declared shelf-life and increase waste. We therefore frame model selection as a decision-support problem: the goal is to prefer structures that are interpretable, stable under decision-relevant evaluation regimes, and sufficiently transparent to be audited across laboratories and QC teams.

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Information criteria such as the Bayesian information criterion (BIC) provide a principled complexity penalty and are known to be consistent under idealized assumptions [1]. However, in small, correlated datasets they can be overly conservative, effectively discouraging structurally meaningful but slightly more complex candidates. Conversely, selecting models solely by cross-validation (CV) can favor higher flexibility for small, noisy error differences, leading to unstable structure selection [2, 3]. This tension is common in food deterioration modeling, where repeated measurements and stage transitions are expected from oxidation chemistry and measurement protocols [6, 7].

To address this gap, we propose **Robustness-Integrated Evidence and Choice (RIEC)**, a mechanism-aware model selection protocol tailored to repeated-measure deterioration data. The acronym highlights three design principles.

Robustness: model conclusions should be stable across evaluation regimes aligned with realistic decision contexts (e.g., extrapolating along time versus generalizing to new product units). We therefore report results under multiple cross-validation (CV) splits and include a random-split negative control to reveal trajectory leakage [8].

Integrated Evidence: we combine parsimony (a BIC-style penalty) with out-of-sample predictive evidence. To avoid overstating information in repeated measures, the complexity penalty is calibrated using an effective sample size n_{eff} derived from the intraclass correlation of trajectories [1, 9].

Choice: we make the selection problem auditable by separating structural assumptions C (e.g., one-phase vs. two-phase dynamics) from within-structure complexity f (e.g., degree, hinge location, or correction flexibility), and by using an explicit two-stage selection procedure.

We demonstrate RIEC on an open olive-oil ageing dataset using UV absorbance indices (K232, K268 and ΔK) that are standard proxies for oxidation and quality change [4–6]. Our goal is not to claim a new chemical mechanism; rather, we provide an audit-ready, decision-aligned selection workflow for repeated-measure food-quality datasets.

Contributions. This paper:

- formalizes the RIEC score that integrates n_{eff}-calibrated parsimony with baseline-normalized out-of-sample evidence;
- shows how evaluation regime choice (time extrapolation vs. oil generalization) affects structure selection, and why random splits are an unreliable baseline for trajectory data;
- provides a reproducible benchmark on olive-oil UV indices, with transparent reporting of structure, complexity, and robustness;
- benchmarks RIEC against common food-modelling selection baselines (AIC, small-sample AICc, and leave-one-unit-out CV) to clarify what is gained by decision-aligned robustness reporting.

2. Materials and methods

2.1. Dataset and targets

We use an open olive-oil ageing dataset released on Mendeley Data [4] and associated methodological analysis [5]. The dataset contains 24 commercial extra virgin olive oils aged at 60 °C with 10 intermediate ageing steps ($n = 240$ UV index rows). For each oil and step, we extract UV absorbance indices K232, K268, and ΔK, following standard UV spectrophotometric methods for olive oil [6]. We analyze each target separately. In the dataset, ageing step is treated as a normalized time variable $t \in [0, 1]$ (scaled from the step index); hinge locations τ are reported on this normalized scale to make comparisons across splits straightforward. K232 and K268 are UV absorbance indices at 232 nm and 268 nm, and ΔK is the IOC-defined spectral deviation index; we use the processed values provided with the dataset and follow the IOC protocol for interpretation [6].

2.2. Notation and key definitions

We index oils by $i = 1, \ldots, N$ (here $N = 24$) and ageing steps by $j = 1, \ldots, m_i$ (here $m_i = 10$), giving $n = \sum_i m_i$ observations. Let $t_{ij} \in [0, 1]$ denote normalized ageing time and y_{ij} denote a target UV index response (K232, K268, or ΔK). A candidate model is denoted by $M = (C, f)$, where C specifies a structural family and f controls within-family complexity. For cross-validation, $\hat{y}_{ij}^{(-k)}$ denotes a prediction produced by a model trained without fold k.

Table 1 summarizes the core quantities used by Robustness-Integrated Evidence and Choice (RIEC).

Table 1
Key notation used in Robustness-Integrated Evidence and Choice (RIEC).

Symbol	Definition
y_{ij}	UV index response for oil i at ageing step j (K232, K268, ΔK).
t_{ij}	Normalized ageing time/step in $[0,1]$ (scaled from the step index).
$M=(C,f)$	Candidate model composed of structure C and complexity choice f within C.
k_M	Number of free parameters in M.
$E_{\mathrm{CV}}(M)$	Cross-validated mean squared prediction error under a chosen regime.
$\mathrm{XPE}(M)$	Relative improvement ratio $E_{\mathrm{CV}}(M_{\mathrm{base}})/E_{\mathrm{CV}}(M)$.
ρ	Intraclass correlation of repeated measures within an oil.
n_{eff}	Effective sample size after design-effect correction.
$C_\lambda(M)$	RIEC score integrating parsimony and out-of-sample evidence.

2.3. Evaluation regimes

To make the conclusions decision-relevant, we define three cross-validation (CV) regimes: (i) **forward-by-time**, where earlier steps are used to predict later steps (forecasting / shelf-life extrapolation); (ii) **group-by-oil**, where entire oils are held out (generalization to new lots/suppliers); and (iii) **random split** as a negative control, which can leak within-oil trajectory information and produce optimistic estimates [8].

In RIEC, **forward-by-time** is treated as the primary regime when the decision question concerns shelf-life extrapolation; **group-by-oil** is used as a robustness check when the question concerns generalization to new oils (new lots, suppliers, or cultivars); and **random split** is included only as a diagnostic negative control. Formally, each regime induces a partition of the index set $\{(i,j)\}$ into training and test folds: forward-by-time uses blocked splits that respect time order, group-by-oil uses grouped splits that hold out entire oils, and random split ignores grouping and therefore can leak trajectory information. We report selection outcomes under all regimes to make robustness (stability of structural choice) explicit.

2.4. Implementation details and recommended reporting

To make the benchmark fully reproducible and to clarify what is being tested by each regime, we summarize implementation details that are often omitted in deterioration modelling papers. Normalized time is computed as $t = \mathrm{ageing\_step}/9$, so that the 10 recorded steps map to $t \in \{0, \frac{1}{9}, \ldots, 1\}$.

Fold design. Forward-by-time uses an *expanding-window* blocked design with $K = 4$ folds: the model is trained on steps 0–1 and tested on 2–3, then trained on 0–3 and tested on 4–5, and so on until testing on the final steps 8–9. Group-by-oil uses $K = 5$ grouped folds that hold out entire oils (4–5 oils per fold), ensuring that each test set corresponds to previously unseen trajectories. Random split uses $K = 5$ folds that ignore oil identity and time order and is therefore treated only as a diagnostic baseline.

Model fitting. Candidates in C_1 and C_2 are linear-in-parameters and are fitted by ordinary least squares. Candidates in C_3 include a nonlinear saturation rate parameter and are fitted by nonlinear least squares with a nonnegativity constraint on the rate parameter to prevent nonphysical growth.

Minimal reporting checklist. For auditability, we recommend reporting: (i) the estimated intraclass correlation ρ and the resulting n_{eff}; (ii) baseline and selected-model E_{CV} and XPE under the primary regime; (iii) the selected structure under at least one alternative, decision-relevant regime; and (iv) whether a random-split negative control produces materially different rankings or suspiciously optimistic errors.

2.5. Cross-validated error and relative improvement (XPE)

For each candidate model M and each evaluation regime, we compute a K-fold cross-validated (CV) mean squared prediction error

$$E_{\mathrm{CV}}(M) = \frac{1}{K}\sum_{k=1}^{K}\frac{1}{|T_k|}\sum_{(i,j)\in T_k}\left(y_{ij} - \hat{y}_{ij}^{(-k)}\right)^2, \tag{1}$$

where T_k is the test set of fold k and $\hat{y}_{ij}^{(-k)}$ denotes a prediction from the model fitted without fold k [2, 3]. We use an intercept-only model M_{base} as a transparent baseline and define the *cross-validated relative improvement ratio*

$$\mathrm{XPE}(M) = \frac{E_{\mathrm{CV}}(M_{\text{base}})}{E_{\mathrm{CV}}(M)}. \tag{2}$$

Thus $\mathrm{XPE}(M) > 1$ indicates improvement over baseline and admits an interpretable error-reduction form: the relative reduction versus baseline is $1 - 1/\mathrm{XPE}(M)$.

2.6. Effective sample size under repeated measures

Repeated measurements within the same oil are correlated, so the nominal sample size n overstates independent information. We quantify within-oil dependence using an intraclass correlation (ICC) ρ and convert it into an effective sample size n_{eff} via a design-effect correction [9].

Concretely, for each target we consider a random-intercept decomposition

$$y_{ij} = \mu + u_i + \varepsilon_{ij}, \qquad u_i \sim \mathcal{N}(0, \sigma_u^2),\ \varepsilon_{ij} \sim \mathcal{N}(0, \sigma_\varepsilon^2), \tag{3}$$

which yields $\rho = \sigma_u^2/(\sigma_u^2 + \sigma_\varepsilon^2)$. Let m_i be the number of repeated measurements for oil i and $\bar{m}$ the average cluster size. We compute the design effect and effective sample size as

$$\mathrm{deff} = 1 + (\bar{m} - 1)\rho, \qquad n_{\text{eff}} = \frac{n}{\mathrm{deff}}. \tag{4}$$

This is the classical Kish design-effect approximation for clustered sampling: it defines n_{eff} as the size of an i.i.d. sample that would yield the same variance inflation induced by within-unit correlation [9]. Because the trajectories here are balanced ($m_i = 10$ for all oils), Eq. (4) is equivalent to the unbalanced-cluster form $n_{\text{eff}} = \sum_i m_i \big/ \big(1 + (m_i - 1)\rho\big)$. We emphasize that n_{eff} is used as a transparent *calibration device* for complexity penalties, not as a substitute for a fully specified longitudinal likelihood. Alternative strategies include computing information criteria from a mixed-effects marginal likelihood (which requires committing to a correlation structure) or using resampling/cluster-bootstrap approaches. RIEC remains agnostic to these modelling choices and only requires a coarse estimate of within-unit dependence to avoid "over-counting" repeated measures.

We use n_{eff} (rather than n) in the complexity-penalty term to avoid overstating evidence for high-flexibility candidates under repeated measures.

2.7. Candidate model family

We enumerate a finite set of nine candidates $M = (C, f)$, where C encodes structural assumptions and f encodes complexity within a structure (Table 2). All candidates take normalized time $t \in [0, 1]$ as input and are fitted by least squares.

Single-phase polynomial (C_1). For degree $d \in \{0, 1, 2\}$,

$$\hat{y}(t) = \beta_0 + \sum_{p=1}^{d} \beta_p t^p. \tag{5}$$

Two-phase hinge / segmented regression (C_2). For a fixed hinge location $\tau \in \{0.4, 0.5, 0.6\}$,

$$\hat{y}(t) = \beta_0 + \beta_1 t + \beta_2 (t - \tau)_+, \qquad (x)_+ = \max(x, 0), \tag{6}$$

which is a continuous piecewise-linear model with a slope change at τ [10].

Mechanism term plus flexible correction (C_3). To allow a simple monotone mechanism shape with residual flexibility, we use a three-parameter saturation term plus a truncated-power correction with $K \in \{1, 2, 3\}$ fixed knots:

$$\hat{y}(t) = \beta_0 + \beta_1 \left(1 - e^{-\beta_2 t}\right) + \sum_{\ell=1}^{K} \gamma_\ell (t - \kappa_\ell)_+^3. \tag{7}$$

Knot locations κ_ℓ are fixed *a priori* (to keep a finite candidate family), so that K controls within-structure flexibility. C_1 (polynomial) includes degrees 0–2; C_2 (two-phase hinge / segmented regression) uses $\tau \in \{0.4, 0.5, 0.6\}$; and C_3 (mechanism + spline-like correction) uses 1–3 knots [10]. For each candidate, we fit by least squares and record the number of free parameters k_M.

Table 2
Candidate model family used in the benchmark (structure C and complexity f).

Model ID	Structure C	Complexity f	Free parameters k_M
C1_poly_deg0	single-phase polynomial	degree 0	1
C1_poly_deg1	single-phase polynomial	degree 1	2
C1_poly_deg2	single-phase polynomial	degree 2	3
C2_hinge_tau0.4	two-phase hinge	$\tau = 0.4$	3
C2_hinge_tau0.5	two-phase hinge	$\tau = 0.5$	3
C2_hinge_tau0.6	two-phase hinge	$\tau = 0.6$	3
C3_mech_spline_k1	mechanism + correction	1 knot	4
C3_mech_spline_k2	mechanism + correction	2 knots	5
C3_mech_spline_k3	mechanism + correction	3 knots	6

2.8. Robustness-Integrated Evidence and Choice (RIEC) score and protocol

Robustness-Integrated Evidence and Choice (RIEC) is a selection protocol designed to make *structure choice* explicit and auditable in repeated-measure settings. It combines three elements.

Robustness. We evaluate candidates under multiple decision-relevant CV regimes (forward-by-time, group-by-oil) and include random split as a negative control. Robustness is reported as the stability of the selected *structure* C across regimes.

Integrated evidence. For a given regime, we combine a parsimony term with out-of-sample evidence. We compute a BIC-style score using n_{eff}:

$$\text{BIC}(M) = -2\log L(\hat{\theta}_M) + k_M \log(n_{\text{eff}}), \tag{8}$$

We compute the Gaussian log-likelihood using the full observation count n (with $\hat{\sigma}^2 = \text{RSS}/n$ under least squares) and replace only the penalty term $\log(n_{\text{eff}})$ to reflect reduced information under within-oil dependence. and form the RIEC score

$$C_\lambda(M) = \text{BIC}(M) - \lambda(n_{\text{eff}})\log\big(\text{XPE}(M)\big), \tag{9}$$

which is equivalent to $\text{BIC}(M) + \lambda(n_{\text{eff}})\log(\text{XPE}(M)^{-1})$. Because $\log(\text{XPE}(M))$ increases with predictive improvement over baseline, the second term rewards out-of-sample evidence while the BIC term penalizes complexity.

We use a decaying weight $\lambda(n_{\text{eff}}) = c/\log(n_{\text{eff}})$ with $c = 2.0$ so that the influence of CV evidence decreases as information grows and the score approaches a pure information criterion in large samples.

Choice. We apply an explicit two-stage selection aligned with the structure/complexity decomposition:

$$f^*(C) = \arg\min_{f\in\mathcal{F}_C} C_\lambda(C, f), \qquad C^* = \arg\min_{C\in\mathcal{C}} C_\lambda\big(C, f^*(C)\big). \tag{10}$$

In the main analysis, C^* is chosen under forward-by-time evaluation (shelf-life extrapolation), and we report whether the same structure is selected under group-by-oil evaluation (robustness to decision context).

Algorithm 1 (RIEC protocol).

1. Specify a finite candidate set $\{M\} = \{(C, f)\}$ and choose one or more evaluation regimes r (forward-by-time; group-by-oil; random split as negative control).
2. For each regime r and each candidate M, compute $E_{\text{CV}}(M; r)$ and $\text{XPE}(M; r)$.
3. Estimate within-oil ICC ρ and compute n_{eff}.
4. Compute $\text{BIC}(M)$ using n_{eff} and then the RIEC score $C_\lambda(M; r)$ via Eq. 9.
5. Perform two-stage choice under the primary regime r_0: pick $f^*(C)$ within each structure, then pick C^* across structures.
6. Report robustness: compare the selected structure across regimes and flag any disagreements.

2.9. Comparator criteria used in food-quality modelling

For context, we also report selections obtained from widely used model-selection baselines in food deterioration studies. These include the Akaike information criterion (AIC) [11], its small-sample correction (AICc) [12], and leave-one-unit-out cross-validation (LOOCV) as an extreme form of grouped CV. For a candidate M with k_M parameters

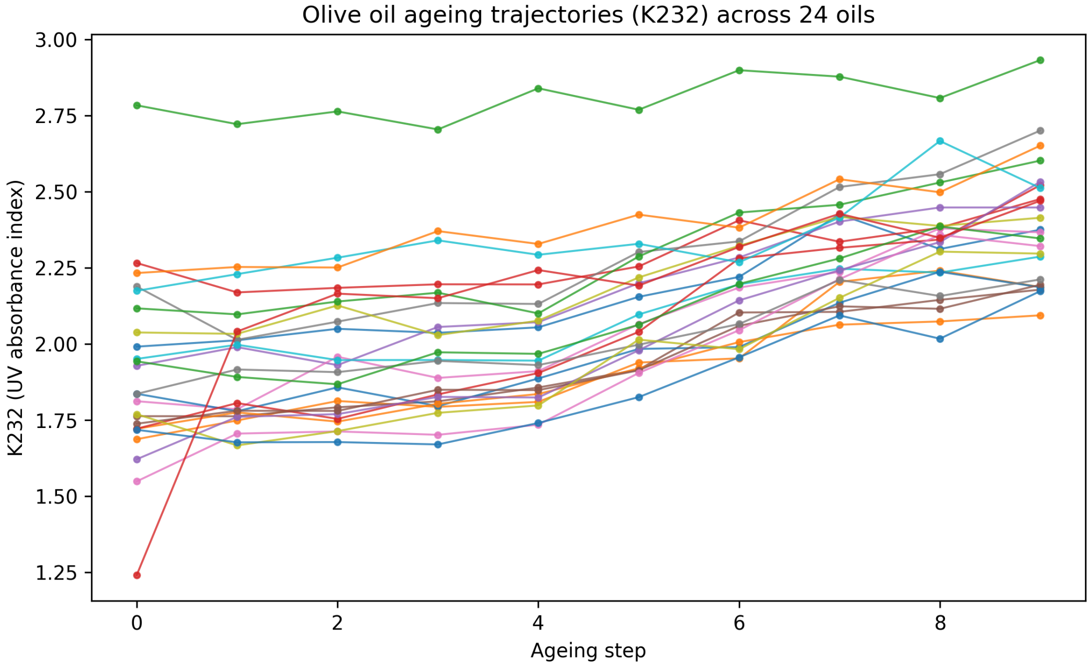

Figure 1: Ageing trajectories of K232 for 24 olive oils across 10 ageing steps. Each line is one oil.

and log-likelihood $\log L(\hat{\theta}_M)$, we define

$$\mathrm{AIC}(M) = -2\log L(\hat{\theta}_M) + 2k_M, \qquad \mathrm{AICc}(M) = \mathrm{AIC}(M) + \frac{2k_M(k_M+1)}{n^\star - k_M - 1}, \tag{11}$$

where $n^\star$ is the information size used for the correction. We report AICc both with $n^\star = n$ (naive) and with $n^\star = n_{\mathrm{eff}}$ as a pragmatic sensitivity analysis for repeated measures. Finally, we compute a leave-one-oil-out CV error $E_{\text{LOO-oil}}$ by holding out each oil trajectory in turn ($K = N$ folds). This grouped LOOCV avoids the row-level leakage that would occur if one left out individual time points within the same trajectory.

3. Results

3.1. Trajectories and effective sample size

Figure 1 shows within-oil trajectories for K232 across ageing steps. The measurement layer estimates within-oil correlation $\rho = 0.500$ and yields an effective sample size $n_{\mathrm{eff}} = 43.6$ (versus nominal $n = 240$), implying that treating points as independent would materially overstate evidence (Figure 2).

3.2. Model selection differs by evaluation regime

Under forward-by-time evaluation (predict later ageing steps from earlier steps), RIEC selects a two-phase hinge model (C_2, $\tau = 0.4$) as the best trade-off between structure and complexity. This model achieves $E_{\mathrm{CV}} = 0.0496$ and XPE$= 1.893$, corresponding to a 47.2% error reduction versus the intercept baseline. Under group-by-oil evaluation (predict unseen oils), the same structure is selected, supporting robustness of the structural choice across decision contexts. Under random split, performance estimates become optimistic and structure ranking can differ, consistent with information leakage in trajectory data (see Supplementary Figure S3). Figure 3 summarizes the selection disagreement under forward-by-time evaluation.

3.3. RIEC stabilizes structure choice versus CV-only

In the group-by-oil regime, CV-only can prefer a more flexible candidate for a marginal 0.41% error gain, while RIEC selects the simpler structured hinge model. The difference is explained by n_{eff}-adjusted complexity penalties

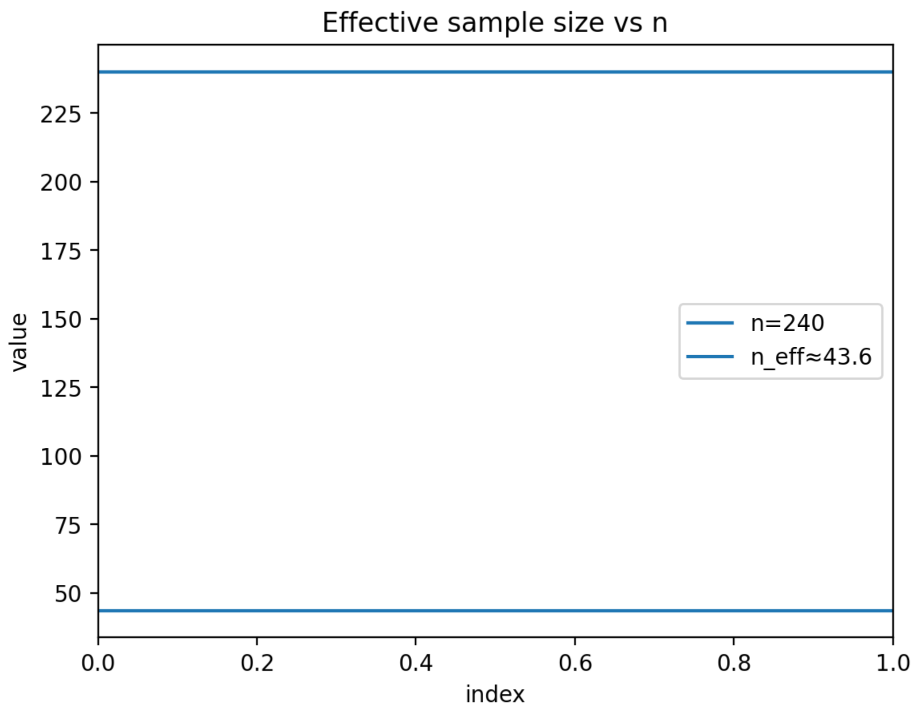

Figure 2: Nominal sample size n vs effective sample size n_{eff} under repeated measures (design-effect correction).

and the calibrated weight on out-of-sample evidence, yielding a more stable and interpretable structural conclusion under limited information.

3.4. Robustness diagnostics: stability across regimes and marginal gains

Table 3 summarizes the primary selection outcomes under each evaluation regime for all three UV-index targets (K232, K268 and ΔK). Under forward-by-time (forecasting), both CV-only and RIEC select the same two-phase hinge model ($\tau = 0.4$), which improves over the intercept baseline by XPE = 1.893 (47.2% error reduction). However, under group-by-oil evaluation (generalization to unseen oils), CV-only selects a more flexible mechanism-plus-correction candidate (C_3, $K = 1$) for a marginal 0.41% error gain relative to the hinge model, illustrating how a CV-only decision can drift toward additional flexibility for small improvements. By contrast, RIEC retains the hinge structure under both regimes because the n_{eff}-calibrated complexity penalty offsets such marginal gains, making the structural conclusion stable.

Rank stability across regimes is quantified in Table 4. For this dataset, the RIEC ranking is identical across the three regimes, whereas CV-only rankings are highly regime dependent (e.g., Spearman $\rho_s = -0.37$ between forward-by-time and group-by-oil ranks). The near-zero correlation between group-by-oil and random-split CV ranks ($\rho \approx 0.02$) further motivates treating random splits as diagnostics rather than as decision-aligned estimates for trajectory data. Across all three targets (K232, K268 and ΔK), RIEC selects the same two-phase hinge model ($\tau = 0.4$) under all three regimes (Table 3).

3.5. Benchmarking against AIC/AICc and leave-one-oil-out CV

To connect RIEC to practices commonly used in food-quality modelling, Table 5 contrasts the top choices from information criteria (BIC, AIC, AICc) and predictive-only criteria (CV-only and LOOCV). A key observation is that naive BIC computed with the nominal n is overly conservative in repeated measures and prefers a single-phase linear trend, whereas using n_{eff} restores preference for the stage-transition hinge structure. Predictive-only criteria can drift toward more flexible correction models when the error gains are extremely small (below 0.5%), which is practically important because such small average-error differences can still translate into noticeably different extrapolations near

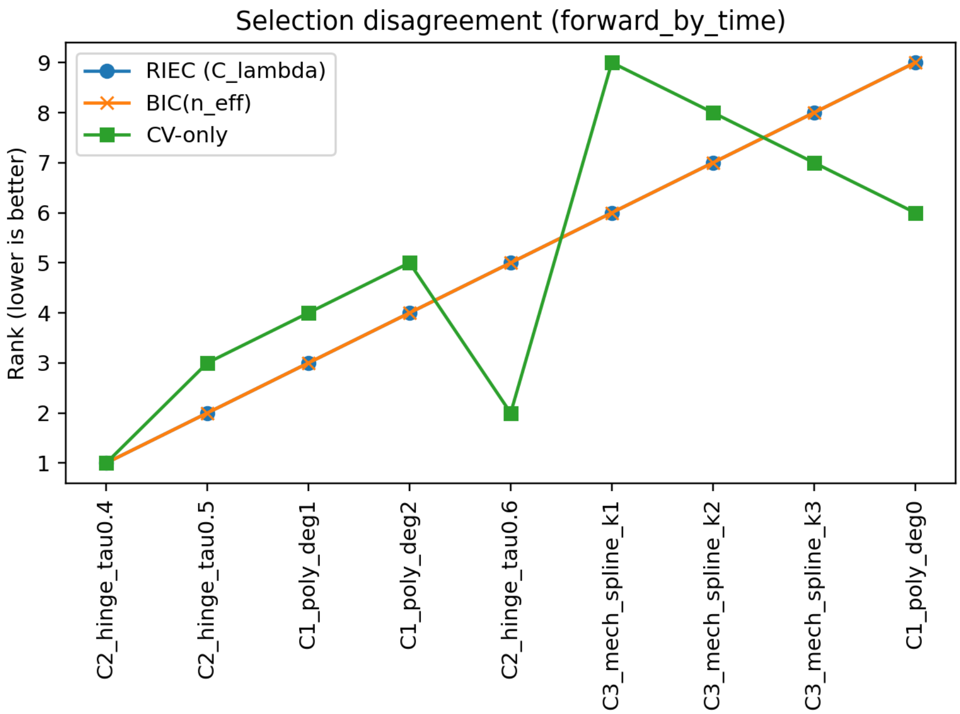

Figure 3: Model selection disagreement under forward-by-time evaluation: RIEC vs CV-only vs BIC(n_{eff}).

QC decision thresholds. RIEC makes this trade-off explicit by combining n_{eff}-calibrated parsimony with decision-aligned evidence and by reporting robustness across regimes.

4. Discussion

4.1. Why a two-phase structure is plausible for oxidation proxies

For edible oils, oxidation chemistry and measurement practice can lead to stage-like behavior (e.g., an initial lag followed by faster change) [7]. In UV spectroscopy of olive oil, K232 and K268 are linked to conjugated diene and triene systems, and ΔK captures local spectral curvature; these indices are routinely used as quality proxies [6]. The hinge structure selected by RIEC captures stage transitions without resorting to high-degree polynomials that are hard to interpret and may extrapolate poorly.

4.2. Why evaluation design matters

Random splits can leak within-oil trajectory information into both train and test sets, producing overly optimistic errors and potentially misleading structure choices. Forward-by-time evaluation aligns with shelf-life and forecasting decisions, while group-by-oil aligns with generalization to new lots or suppliers. These concerns mirror established cautions about using standard CV blindly on dependent or ordered data [8].

4.3. What RIEC adds beyond standard practices

BIC provides an explicit complexity penalty but can be miscalibrated when the effective information is much smaller than n due to repeated measures [1, 9]. CV directly estimates out-of-sample performance but can be high-variance in small samples and may prefer extra flexibility for tiny error differences [2, 3]. RIEC operationalizes a practical bridge: it adjusts information content via n_{eff}, and it uses a decaying weight so that predictive evidence has more influence when information is scarce but fades as information grows.

Table 3
Selections and performance across evaluation regimes for each target UV index (K232, K268 and ΔK). Effective sample sizes used in RIEC were target-specific (K232: $n_{\text{eff}} = 43.6$; K268: $n_{\text{eff}} = 105.7$; ΔK: $n_{\text{eff}} = 167.6$).

Panel A: Baseline and RIEC.

Regime	Baseline E_{CV}	RIEC choice	E_{CV}	XPE
K232				
Forward-by-time ($K = 4$)	0.0939	C_2 hinge ($\tau = 0.4$)	0.0496	1.893
Group-by-oil ($K = 5$)	0.0817	C_2 hinge ($\tau = 0.4$)	0.0521	1.568
Random split ($K = 5$)	0.0811	C_2 hinge ($\tau = 0.4$)	0.0525	1.544
K268				
Forward-by-time ($K = 4$)	0.0081	C_2 hinge ($\tau = 0.4$)	0.0024	3.316
Group-by-oil ($K = 5$)	0.0057	C_2 hinge ($\tau = 0.4$)	0.0018	3.128
Random split ($K = 5$)	0.0056	C_2 hinge ($\tau = 0.4$)	0.0017	3.264
ΔK				
Forward-by-time ($K = 4$)	5.52×10^{-5}	C_2 hinge ($\tau = 0.4$)	1.12×10^{-5}	4.927
Group-by-oil ($K = 5$)	3.66×10^{-5}	C_2 hinge ($\tau = 0.4$)	7.76×10^{-6}	4.710
Random split ($K = 5$)	3.65×10^{-5}	C_2 hinge ($\tau = 0.4$)	7.79×10^{-6}	4.691

Panel B: CV-only (predictive-only selection).

Regime	CV-only choice	E_{CV}
K232		
Forward-by-time ($K = 4$)	C_2 hinge ($\tau = 0.4$)	0.0496
Group-by-oil ($K = 5$)	C_3 mech+spline ($K = 1$)	0.0519
Random split ($K = 5$)	C_2 hinge ($\tau = 0.4$)	0.0525
K268		
Forward-by-time ($K = 4$)	C_2 hinge ($\tau = 0.4$)	0.0024
Group-by-oil ($K = 5$)	C_3 mech+spline ($K = 3$)	0.0018
Random split ($K = 5$)	C_2 hinge ($\tau = 0.4$)	0.0017
ΔK		
Forward-by-time ($K = 4$)	C_2 hinge ($\tau = 0.4$)	1.12×10^{-5}
Group-by-oil ($K = 5$)	C_3 mech+spline ($K = 3$)	7.62×10^{-6}
Random split ($K = 5$)	C_2 hinge ($\tau = 0.4$)	7.79×10^{-6}

Table 4
Rank stability across regimes (Spearman correlation of candidate ranks). For CV-only, ranks are based on E_{CV}; for BIC(n_{eff}) on the information criterion; and for RIEC on C_λ.

Method	Forward vs group	Forward vs random	Group vs random
CV-only	-0.37	0.73	0.02
BIC(n_{eff})	1.00	1.00	1.00
RIEC	1.00	1.00	1.00

4.4. Food-industry value: QC-relevant model choice and auditability

In many food applications, the end user of a deterioration model is a QC scientist or process engineer who must justify shelf-life labels and compliance decisions. Even when average prediction errors differ only marginally, extrapolated trajectories can cross acceptance thresholds at noticeably different times, especially near a phase transition. For olive oil, UV indices are part of routine compliance screening [6]; in this setting, selecting an over-flexible curve can "chase" within-trajectory noise and shift the implied time-to-limit. RIEC is designed to reduce this risk pragmatically: it prefers interpretable structures unless there is clear out-of-sample evidence for extra flexibility, and it requires that conclusions be reported under decision-relevant evaluation regimes.

Table 5
Comparison with common selection baselines for K232. BIC(n) uses the nominal sample size, while BIC(n_{eff}) uses the effective sample size from Eq. (4). AIC and AICc are computed from the Gaussian likelihood of each candidate; AICc is reported with $n^\star = n$ and with $n^\star = n_{\text{eff}}$. LOOCV is implemented as leave-one-oil-out grouped CV.

Criterion	Selected model	Metric / comment
RIEC (primary: forward-by-time)	C2 hinge ($\tau = 0.4$)	$E_{\text{CV}} = 0.0496$; structure remains selected under group-by-oil.
BIC(n) (naive)	C1 polynomial (deg 1)	Overly conservative under repeated measures; forward-by-time $E_{\text{CV}} = 0.0503$ (1.44% worse than hinge).
BIC(n_{eff})	C2 hinge ($\tau = 0.4$)	Dependence-corrected complexity restores the stage-transition structure.
AIC / AICc(n) / AICc(n_{eff})	C2 hinge ($\tau = 0.4$)	Same top choice here, but these criteria do not encode decision-aligned CV regimes or leakage diagnostics.
CV-only (group-by-oil, $K = 5$)	C3 mech+spline ($K = 1$)	$E_{\text{CV}} = 0.0519$, a marginal 0.41% gain vs hinge, illustrating drift to extra flexibility for tiny gains.
LOOCV (leave-one-oil-out, $K = N$)	C3 mech+spline ($K = 3$)	$E_{\text{LOO-oil}} = 0.0552$, a marginal 0.28% gain vs hinge; differences are tiny but the selected model is more complex.

4.5. Practical guidance for applying RIEC in food-quality studies

While RIEC is presented here as a benchmark protocol, it can be used as a pragmatic "decision wrapper" around standard candidate modelling. In practice, we recommend the following workflow for repeated-measure deterioration datasets:

- **Define the decision context first.** If the goal is shelf-life extrapolation, use a forward-by-time regime as the primary target; if the goal is supplier/lot transfer, use group-by-unit as the primary target.
- **Keep the candidate set finite and interpretable.** Include a small number of mechanistically plausible structures (e.g., one-phase vs. two-phase) and a limited set of within-structure flexibilities so that selection outcomes are auditable rather than purely data-driven.
- **Report dependence and effective information.** Estimate ρ and n_{eff} and explicitly state whether complexity penalties are calibrated to n or to n_{eff}.
- **Use random splits only diagnostically.** If random-split errors are much lower than blocked or grouped errors, or if rankings change drastically, this is evidence that naive CV may be misaligned with the intended decision.

These steps aim to prevent overconfident conclusions from trajectory leakage and to make the rationale for model choice transparent to reviewers and practitioners.

4.6. Limitations and next steps

This benchmark uses UV proxies rather than full chemical panels (e.g., peroxide value, acid value, volatile markers, or sensory scores), and ageing is accelerated at 60 °C [4, 5]. While we analyse three UV indices (K232, K268 and ΔK), all originate from the same measurement modality; direct translation to ambient shelf-life would require additional datasets and, ideally, multi-temperature models (e.g., Arrhenius-type extensions). The RIEC protocol itself is general: it applies to any repeated-measure deterioration dataset once the dependence "unit" (cluster) and the decision context (forecasting along time vs. generalization to new units) are specified. We used a Kish design-effect correction for a transparent n_{eff}; future work should compare this calibration to likelihood-based mixed-effects information criteria and to resampling-based uncertainty estimates on richer and more unbalanced longitudinal datasets. Finally, broader empirical validation on other food systems (other edible oils, dairy, meat, plant-based products) and other indicator types would strengthen evidence for general applicability and help establish community reporting norms.

5. Conclusions

In a public olive-oil UV ageing dataset with repeated measures, treating rows as independent can inflate evidence and destabilize model choice. RIEC provides a mechanism-aware, audit-ready model selection workflow: it separates structure from complexity, corrects information content via an effective sample size, and combines parsimony with out-of-sample evidence. Across decision-aligned evaluation regimes, RIEC identifies a simple two-phase hinge structure that yields strong forecasting performance while remaining interpretable for food-quality practice.

Data and code availability

Data: The study uses a public dataset available on Mendeley Data (DOI: 10.17632/g6y69g8gwm.1) [4]. Code: A runnable reproduction package is available at `https://github.com/lijincheng494gmail/crfs-reproduce` (archive: `https://github.com/lijincheng494gmail/crfs-reproduce/raw/main/riec_crfs_reproduce.zip`). A brief runbook with one-click commands and expected outputs is provided in the submission files (see the Reproducibility links & instructions document).

CRediT authorship contribution statement

Jincheng Li: Modelling experiments, Methodology, Software, Formal analysis, Writing—original draft, Writing—review & editing. Xifeng Li and Jinlong Ji: Domain guidance and calibration (food and edible oils), Validation (domain applicability check), Writing—review & editing.

Declaration of competing interest

The authors declare no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Jincheng Li: Modelling experiments; Methodology; Software; Formal analysis; Writing—original draft; Writing—review & editing.

Xifeng Li: Domain guidance and calibration (food and edible oils); Validation (domain applicability check); Writing—review & editing.

Jinlong Ji: Domain guidance and calibration (food and edible oils); Validation (domain applicability check); Writing—review & editing.

Declaration of Competing Interests

Journal: Current Research in Food Science

Manuscript title: Mechanism-aware model selection for food-quality deterioration under repeated measures: a Robustness-Integrated Evidence and Choice (RIEC) benchmark on olive oil UV indices

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The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Signed on behalf of all authors,

Jincheng Li

2026-02-24

Table(Editable version)

Target	Regime	Baseline_E	RIEC_choic	RIEC_E_CV	XPE
K232	Forward-by	0.0939	C2 hinge (ta	0.0496	1.893
K232	Group-by-c	0.0817	C2 hinge (ta	0.0521	1.568
K232	Random sp	0.0811	C2 hinge (ta	0.0525	1.544
K268	Forward-by	0.0081	C2 hinge (ta	0.0024	3.316
K268	Group-by-c	0.0057	C2 hinge (ta	0.0018	3.128
K268	Random sp	0.0056	C2 hinge (ta	0.0017	3.264
DeltaK	Forward-by	5.52E-05	C2 hinge (ta	1.12E-05	4.927
DeltaK	Group-by-c	3.66E-05	C2 hinge (ta	7.76E-06	4.71
DeltaK	Random sp	3.65E-05	C2 hinge (ta	7.79E-06	4.691

Target	Regime	CV_only_cl	CV_only_E_CV
K232	Forward-by	C2 hinge (ta	0.0496
K232	Group-by-c	C3 mech+s	0.0519
K232	Random sp	C2 hinge (ta	0.0525
K268	Forward-by	C2 hinge (ta	0.0024
K268	Group-by-c	C3 mech+s	0.0018
K268	Random sp	C2 hinge (ta	0.0017
DeltaK	Forward-by	C2 hinge (ta	1.12E-05
DeltaK	Group-by-c	C3 mech+s	7.62E-06
DeltaK	Random sp	C2 hinge (ta	7.79E-06

Supplementary File 2
Mathematical Definitions and Derivations

(for the manuscript "Robustness-Integrated Evidence and Choice (RIEC)")

1 Purpose

This note provides formal definitions and compact derivations for the key quantities used in the **Robustness-Integrated Evidence and Choice (RIEC)** protocol. It is designed to make the manuscript mathematically self-contained for statistical / modelling reviewers.

2 Notation

2.1 Data structure

Let $i \in \{1, \dots, N\}$ index oils (clusters) and $j \in \{0, \dots, 9\}$ index ageing steps. For a given target (K232, K268, or ΔK), let y_{ij} denote the observed value at step j for oil i. Let $t_{ij} \in [0, 1]$ denote normalized time:

$$t_{ij} = \frac{j}{9} = \frac{\text{ageing\_step}}{9}.$$

The total number of rows is $n = Nm$ where $m = 10$ in this benchmark (balanced repeats).

2.2 Candidate family

A candidate model is denoted by $M = (C, f)$:

- C is the *structural form* (e.g., single-phase polynomial vs two-phase hinge vs mechanism-plus-spline).
- f indexes within-structure complexity (e.g., polynomial degree, hinge location τ, number of spline knots).
- k_M is the number of free parameters in the fitted model.

Let M_0 denote the baseline intercept-only model.

2.3 Cross-validation folds

For a fixed evaluation regime, let $\{T_k\}_{k=1}^{K}$ denote the K test sets (folds), where T_k is a set of indices (i, j). Let $\widehat{y}_{ij}^{(-k)}(M)$ denote the fitted prediction for $(i, j) \in T_k$ produced by training M on the corresponding training set (all indices not in T_k, subject to the regime constraints).

3 Cross-validated error

For each candidate M, the protocol computes a mean squared prediction error averaged across folds:

$$E_{\text{CV}}(M) = \frac{1}{K} \sum_{k=1}^{K} \frac{1}{|T_k|} \sum_{(i,j) \in T_k} \left(y_{ij} - \widehat{y}_{ij}^{(-k)}(M) \right)^2. \tag{1}$$

This definition is regime-agnostic: forward-by-time, group-by-oil, and random split correspond to different constructions of the test sets T_k.

Remark (why fold-level normalization matters). We normalize by $|T_k|$ inside each fold so that each fold contributes equally to the final score, even when test folds have slightly different sizes.

4 Relative improvement ratio (XPE)

RIEC uses a unit-free improvement ratio to compare predictive gains over the baseline:

$$\mathrm{XPE}(M) = \frac{E_{\mathrm{CV}}(M_0)}{E_{\mathrm{CV}}(M)}. \tag{2}$$

If $\mathrm{XPE}(M) > 1$, then M improves over the baseline (lower CV error). A convenient interpretation is the relative error reduction:

$$\frac{E_{\mathrm{CV}}(M_0) - E_{\mathrm{CV}}(M)}{E_{\mathrm{CV}}(M_0)} = 1 - \frac{E_{\mathrm{CV}}(M)}{E_{\mathrm{CV}}(M_0)} = 1 - \frac{1}{\mathrm{XPE}(M)}.$$

Thus, for example, $\mathrm{XPE} = 1.893$ corresponds to an error reduction of $1 - 1/1.893 \approx 47.2\%$.

5 Effective sample size under repeated measures

5.1 Random-intercept working model and ICC

To quantify within-oil dependence, consider the random-intercept model

$$y_{ij} = \mu + u_i + \varepsilon_{ij}, \tag{3}$$

where $u_i \sim (0, \sigma_u^2)$ represents oil-to-oil variation and $\varepsilon_{ij} \sim (0, \sigma_\varepsilon^2)$ is an idiosyncratic error. The intraclass correlation (ICC) is

$$\rho = \frac{\sigma_u^2}{\sigma_u^2 + \sigma_\varepsilon^2}. \tag{4}$$

Under (3), any two observations from the same oil have correlation ρ.

5.2 Design effect and n_{eff} (balanced case)

Assume each oil contributes m repeated measurements (balanced trajectory length), and define the overall sample mean $\bar{y} = \frac{1}{n} \sum_{i=1}^{N} \sum_{j=1}^{m} y_{ij}$. Let $\sigma^2 = \sigma_u^2 + \sigma_\varepsilon^2$. A standard calculation gives

$$\mathrm{Var}(\bar{y}) = \frac{\sigma^2}{n}\Big(1 + (m-1)\rho\Big).$$

Derivation sketch. Write $\bar{y}$ as the average of all y_{ij}, expand $\mathrm{Var}(\bar{y})$ using the within-oil covariances $\mathrm{Cov}(y_{ij}, y_{ij'}) = \rho\sigma^2$ for $j \neq j'$ and 0 across oils, and count the number of within-oil pairs.

The multiplicative inflation factor

$$\mathrm{deff} = 1 + (m-1)\rho \tag{5}$$

is the *design effect*. It motivates the effective sample size

$$n_{\mathrm{eff}} = \frac{n}{\mathrm{deff}} = \frac{n}{1 + (m-1)\rho}. \tag{6}$$

Intuitively, n_{eff} is the number of independent observations that would yield the same information content (for estimating a mean) as the correlated repeated-measure sample.

5.3 Unbalanced extension (optional)

If cluster sizes differ (m_i varies by oil), one common approximation replaces m in (5) by

$$m^\star = \frac{\sum_{i=1}^{N} m_i^2}{\sum_{i=1}^{N} m_i}, \quad \text{giving} \quad \text{deff} \approx 1 + (m^\star - 1)\rho.$$

In this benchmark, trajectories are balanced ($m_i \equiv 10$), so $m^\star = \bar{m} = 10$.

Important scope note. RIEC uses n_{eff} to *calibrate complexity penalties* under repeated measures. Cross-validated prediction errors E_{CV} are still computed on the original data and the decision-aligned folds.

6 Information criterion with repeated-measure calibration

For a candidate M with maximized likelihood $L(\widehat{\theta}_M)$, define a BIC-style score

$$\text{BIC}(M) = -2 \log L(\widehat{\theta}_M) + k_M \log(n_{\text{eff}}). \tag{7}$$

Replacing n by n_{eff} increases the penalty when within-oil dependence is strong (small n_{eff}), discouraging over-flexible models that exploit within-trajectory correlation.

7 RIEC score

7.1 Definition and equivalent forms

RIEC combines parsimony and predictive evidence via

$$C_\lambda(M) = \text{BIC}(M) - \lambda(n_{\text{eff}}) \log \big(\text{XPE}(M)\big). \tag{8}$$

Because $\log(\text{XPE}(M)) > 0$ when M improves over baseline, the second term reduces the score for predictively better candidates. An equivalent form is

$$C_\lambda(M) = \text{BIC}(M) + \lambda(n_{\text{eff}}) \log \left(\text{XPE}(M)^{-1}\right).$$

7.2 Choice of $\lambda(n_{\text{eff}})$

We use a decaying weight

$$\lambda(n_{\text{eff}}) = \frac{c}{\log(n_{\text{eff}})}, \tag{9}$$

with $c = 2.0$ in the benchmark. This choice has two practical properties:

- When effective information is scarce (small n_{eff}), λ is larger and predictive evidence is weighted more heavily.
- As $n_{\text{eff}} \to \infty$, $\lambda(n_{\text{eff}}) \to 0$, so C_λ approaches the BIC term.

8 Two-stage selection rule (separating structure vs complexity)

Let $\mathcal{C}$ denote the finite set of structural forms and $\mathcal{F}_C$ the set of allowable complexity settings under structure C. RIEC performs:

1. **Within-structure selection:** for each $C \in \mathcal{C}$, pick

$$f_C^\star = \arg \min_{f \in \mathcal{F}_C} C_\lambda(C, f).$$

2. **Across-structure choice:** select the structure

$$C^\star = \arg\min_{C\in\mathcal{C}} C_\lambda\big(C, f_C^\star\big),$$

and report $M^\star = (C^\star, f_{C^\star}^\star)$.

This two-stage form makes the selection auditable: readers can see whether disagreement comes from (i) a different structural worldview or (ii) different within-structure tuning.

9 Robustness via regime comparison and negative controls

RIEC reports selection outcomes under multiple evaluation regimes. In repeated-measure deterioration settings, a key diagnostic is whether random row-level splits yield suspiciously optimistic errors and unstable rankings relative to group-aware regimes. Large gaps between random-split performance and group-by-oil performance can signal *trajectory leakage* (train-test dependence).

Mechanism-aware model selection for food-quality deterioration under repeated measures: a Robustness-Integrated Evidence and Choice (RIEC) benchmark on olive oil UV indices

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ABSTRACT

Food-quality deterioration datasets increasingly contain repeated measurements for each product unit, forming correlated within-unit trajectories and often exhibiting stage transitions. These properties complicate model selection: information criteria can miscalibrate complexity when observations are dependent, while cross-validation can prefer over-flexible candidates for small error gains. We present **Robustness-Integrated Evidence and Choice (RIEC)**, a mechanism-aware model selection protocol for finite candidate families. RIEC (i) evaluates candidates under decision-relevant cross-validation regimes and uses a random-split negative control to diagnose trajectory leakage (*Robustness*), (ii) integrates a BIC-style parsimony term based on an effective sample size with an out-of-sample evidence term derived from baseline-normalized CV error (*Evidence*), and (iii) performs an explicit two-stage selection that separates structural assumptions from within-structure complexity (*Choice*).

Using an open olive-oil ageing dataset (24 oils, 10 ageing steps; $n = 240$), we model UV absorbance indices (K232, K268 and ΔK) as oxidation proxies. We design three evaluation regimes to reflect distinct decision contexts: forward-by-time extrapolation, group-by-oil generalization, and a random split as a negative control. The measurement layer estimates strong within-oil dependence ($\rho = 0.500$) and an effective sample size ($n_{\mathrm{eff}} = 43.6$), which materially changes complexity penalties under repeated measures. Under forward-by-time evaluation, RIEC selects a two-phase hinge model ($\tau = 0.4$) and achieves $E_{\mathrm{CV}} = 0.0496$ with XPE $= 1.893$, corresponding to a 47.2% error reduction versus an intercept baseline. Structural choice is consistent under group-by-oil evaluation, whereas random splits yield optimistic errors and unstable rankings. RIEC provides an audit-ready workflow for repeated-measure deterioration modelling without claiming a new chemical mechanism.

1. Introduction

Accurate modeling of food-quality deterioration is central to quality control, process optimization, and shelf-life management. In lipid-rich products such as edible oils, oxidation-related indicators often evolve nonlinearly and may exhibit a lag phase followed by acceleration [?]. Public datasets increasingly provide repeated measurements over time for each product unit, forming within-unit trajectories rather than independent observations [? ?]. When trajectories are treated as independent samples, both uncertainty quantification and complexity penalties can be miscalibrated. Model selection becomes particularly sensitive when researchers must choose between mechanistic structures and flexible empirical fits.

In food-industry settings, the cost of model misspecification is rarely summarized by a single average error metric. Quality control (QC) decisions often depend on predicted times to cross regulatory or internal acceptance thresholds (e.g., IOC UV limits for extra virgin olive oil) and on whether a model extrapolates conservatively under limited information [?]. Over-flexible candidates can absorb within-unit noise and shift the estimated "time-to-limit", which may increase the risk of shelf-life mislabelling, avoidable product holds/recalls, or non-compliance, while overly conservative choices can shorten declared shelf-life and increase waste. We therefore frame model selection as a decision-support problem: the goal is to prefer structures that are interpretable, stable under decision-relevant evaluation regimes, and sufficiently transparent to be audited across laboratories and QC teams.

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Information criteria such as the Bayesian information criterion (BIC) provide a principled complexity penalty and are known to be consistent under idealized assumptions [**?**]. However, in small, correlated datasets they can be overly conservative, effectively discouraging structurally meaningful but slightly more complex candidates. Conversely, selecting models solely by cross-validation (CV) can favor higher flexibility for small, noisy error differences, leading to unstable structure selection [**? ?**]. This tension is common in food deterioration modeling, where repeated measurements and stage transitions are expected from oxidation chemistry and measurement protocols [**? ?**].

To address this gap, we propose **Robustness-Integrated Evidence and Choice (RIEC)**, a mechanism-aware model selection protocol tailored to repeated-measure deterioration data. The acronym highlights three design principles.

Robustness: model conclusions should be stable across evaluation regimes aligned with realistic decision contexts (e.g., extrapolating along time versus generalizing to new product units). We therefore report results under multiple cross-validation (CV) splits and include a random-split negative control to reveal trajectory leakage [**?**].

Integrated Evidence: we combine parsimony (a BIC-style penalty) with out-of-sample predictive evidence. To avoid overstating information in repeated measures, the complexity penalty is calibrated using an effective sample size n_{eff} derived from the intraclass correlation of trajectories [**? ?**].

Choice: we make the selection problem auditable by separating structural assumptions C (e.g., one-phase vs. two-phase dynamics) from within-structure complexity f (e.g., degree, hinge location, or correction flexibility), and by using an explicit two-stage selection procedure.

We demonstrate RIEC on an open olive-oil ageing dataset using UV absorbance indices (K232, K268 and ΔK) that are standard proxies for oxidation and quality change [**? ? ?**]. Our goal is not to claim a new chemical mechanism; rather, we provide an audit-ready, decision-aligned selection workflow for repeated-measure food-quality datasets.

Contributions. This paper:

- formalizes the RIEC score that integrates n_{eff}-calibrated parsimony with baseline-normalized out-of-sample evidence;
- shows how evaluation regime choice (time extrapolation vs. oil generalization) affects structure selection, and why random splits are an unreliable baseline for trajectory data;
- provides a reproducible benchmark on olive-oil UV indices, with transparent reporting of structure, complexity, and robustness;
- benchmarks RIEC against common food-modelling selection baselines (AIC, small-sample AICc, and leave-one-unit-out CV) to clarify what is gained by decision-aligned robustness reporting.

2. Materials and methods

2.1. Dataset and targets

We use an open olive-oil ageing dataset released on Mendeley Data [**?**] and associated methodological analysis [**?**]. The dataset contains 24 commercial extra virgin olive oils aged at 60 °C with 10 intermediate ageing steps ($n = 240$ UV index rows). For each oil and step, we extract UV absorbance indices K232, K268, and ΔK, following standard UV spectrophotometric methods for olive oil [**?**]. We analyze each target separately. In the dataset, ageing step is treated as a normalized time variable $t \in [0, 1]$ (scaled from the step index); hinge locations τ are reported on this normalized scale to make comparisons across splits straightforward. K232 and K268 are UV absorbance indices at 232 nm and 268 nm, and ΔK is the IOC-defined spectral deviation index; we use the processed values provided with the dataset and follow the IOC protocol for interpretation [**?**].

2.2. Notation and key definitions

We index oils by $i = 1, \ldots, N$ (here $N = 24$) and ageing steps by $j = 1, \ldots, m_i$ (here $m_i = 10$), giving $n = \sum_i m_i$ observations. Let $t_{ij} \in [0, 1]$ denote normalized ageing time and y_{ij} denote a target UV index response (K232, K268, or ΔK). A candidate model is denoted by $M = (C, f)$, where C specifies a structural family and f controls within-family complexity. For cross-validation, $\hat{y}_{ij}^{(-k)}$ denotes a prediction produced by a model trained without fold k.

Table **??** summarizes the core quantities used by Robustness-Integrated Evidence and Choice (RIEC).

Table 1
Key notation used in Robustness-Integrated Evidence and Choice (RIEC).

Symbol	Definition
y_{ij}	UV index response for oil i at ageing step j (K232, K268, ΔK).
t_{ij}	Normalized ageing time/step in $[0, 1]$ (scaled from the step index).
$M = (C, f)$	Candidate model composed of structure C and complexity choice f within C.
k_M	Number of free parameters in M.
$E_{\mathrm{CV}}(M)$	Cross-validated mean squared prediction error under a chosen regime.
$\mathrm{XPE}(M)$	Relative improvement ratio $E_{\mathrm{CV}}(M_{\mathrm{base}})/E_{\mathrm{CV}}(M)$.
ρ	Intraclass correlation of repeated measures within an oil.
n_{eff}	Effective sample size after design-effect correction.
$C_\lambda(M)$	RIEC score integrating parsimony and out-of-sample evidence.

2.3. Evaluation regimes

To make the conclusions decision-relevant, we define three cross-validation (CV) regimes: (i) **forward-by-time**, where earlier steps are used to predict later steps (forecasting / shelf-life extrapolation); (ii) **group-by-oil**, where entire oils are held out (generalization to new lots/suppliers); and (iii) **random split** as a negative control, which can leak within-oil trajectory information and produce optimistic estimates [**?**].

In RIEC, **forward-by-time** is treated as the primary regime when the decision question concerns shelf-life extrapolation; **group-by-oil** is used as a robustness check when the question concerns generalization to new oils (new lots, suppliers, or cultivars); and **random split** is included only as a diagnostic negative control. Formally, each regime induces a partition of the index set $\{(i, j)\}$ into training and test folds: forward-by-time uses blocked splits that respect time order, group-by-oil uses grouped splits that hold out entire oils, and random split ignores grouping and therefore can leak trajectory information. We report selection outcomes under all regimes to make robustness (stability of structural choice) explicit.

2.4. Implementation details and recommended reporting

To make the benchmark fully reproducible and to clarify what is being tested by each regime, we summarize implementation details that are often omitted in deterioration modelling papers. Normalized time is computed as $t = \text{ageing\_step}/9$, so that the 10 recorded steps map to $t \in \{0, \frac{1}{9}, \dots, 1\}$.

Fold design. Forward-by-time uses an *expanding-window* blocked design with $K = 4$ folds: the model is trained on steps 0–1 and tested on 2–3, then trained on 0–3 and tested on 4–5, and so on until testing on the final steps 8–9. Group-by-oil uses $K = 5$ grouped folds that hold out entire oils (4–5 oils per fold), ensuring that each test set corresponds to previously unseen trajectories. Random split uses $K = 5$ folds that ignore oil identity and time order and is therefore treated only as a diagnostic baseline.

Model fitting. Candidates in C_1 and C_2 are linear-in-parameters and are fitted by ordinary least squares. Candidates in C_3 include a nonlinear saturation rate parameter and are fitted by nonlinear least squares with a nonnegativity constraint on the rate parameter to prevent nonphysical growth.

Minimal reporting checklist. For auditability, we recommend reporting: (i) the estimated intraclass correlation ρ and the resulting n_{eff}; (ii) baseline and selected-model E_{CV} and XPE under the primary regime; (iii) the selected structure under at least one alternative, decision-relevant regime; and (iv) whether a random-split negative control produces materially different rankings or suspiciously optimistic errors.

2.5. Cross-validated error and relative improvement (XPE)

For each candidate model M and each evaluation regime, we compute a K-fold cross-validated (CV) mean squared prediction error

$$E_{\mathrm{CV}}(M) = \frac{1}{K} \sum_{k=1}^{K} \frac{1}{|T_k|} \sum_{(i,j)\in T_k} \left(y_{ij} - \hat{y}_{ij}^{(-k)} \right)^2, \tag{1}$$

where T_k is the test set of fold k and $\hat{y}_{ij}^{(-k)}$ denotes a prediction from the model fitted without fold k [**?** **?**]. We use an intercept-only model M_{base} as a transparent baseline and define the *cross-validated relative improvement ratio*

$$\text{XPE}(M) = \frac{E_{\text{CV}}(M_{\text{base}})}{E_{\text{CV}}(M)}. \tag{2}$$

Thus $\text{XPE}(M) > 1$ indicates improvement over baseline and admits an interpretable error-reduction form: the relative reduction versus baseline is $1 - 1/\text{XPE}(M)$.

2.6. Effective sample size under repeated measures

Repeated measurements within the same oil are correlated, so the nominal sample size n overstates independent information. We quantify within-oil dependence using an intraclass correlation (ICC) ρ and convert it into an effective sample size n_{eff} via a design-effect correction [**?**].

Concretely, for each target we consider a random-intercept decomposition

$$y_{ij} = \mu + u_i + \varepsilon_{ij}, \qquad u_i \sim \mathcal{N}(0, \sigma_u^2),\ \varepsilon_{ij} \sim \mathcal{N}(0, \sigma_\varepsilon^2), \tag{3}$$

which yields $\rho = \sigma_u^2/(\sigma_u^2 + \sigma_\varepsilon^2)$. Let m_i be the number of repeated measurements for oil i and $\bar{m}$ the average cluster size. We compute the design effect and effective sample size as

$$\text{deff} = 1 + (\bar{m} - 1)\rho, \qquad n_{\text{eff}} = \frac{n}{\text{deff}}. \tag{4}$$

This is the classical Kish design-effect approximation for clustered sampling: it defines n_{eff} as the size of an i.i.d. sample that would yield the same variance inflation induced by within-unit correlation [**?**]. Because the trajectories here are balanced ($m_i = 10$ for all oils), Eq. (**??**) is equivalent to the unbalanced-cluster form $n_{\text{eff}} = \sum_i m_i \big/ \big(1 + (m_i - 1)\rho\big)$. We emphasize that n_{eff} is used as a transparent *calibration device* for complexity penalties, not as a substitute for a fully specified longitudinal likelihood. Alternative strategies include computing information criteria from a mixed-effects marginal likelihood (which requires committing to a correlation structure) or using resampling/cluster-bootstrap approaches. RIEC remains agnostic to these modelling choices and only requires a coarse estimate of within-unit dependence to avoid "over-counting" repeated measures.

We use n_{eff} (rather than n) in the complexity-penalty term to avoid overstating evidence for high-flexibility candidates under repeated measures.

2.7. Candidate model family

We enumerate a finite set of nine candidates $M = (C, f)$, where C encodes structural assumptions and f encodes complexity within a structure (Table **??**). All candidates take normalized time $t \in [0, 1]$ as input and are fitted by least squares.

Single-phase polynomial (C_1). For degree $d \in \{0, 1, 2\}$,

$$\hat{y}(t) = \beta_0 + \sum_{p=1}^{d} \beta_p t^p. \tag{5}$$

Two-phase hinge / segmented regression (C_2). For a fixed hinge location $\tau \in \{0.4, 0.5, 0.6\}$,

$$\hat{y}(t) = \beta_0 + \beta_1 t + \beta_2 (t - \tau)_+, \qquad (x)_+ = \max(x, 0), \tag{6}$$

which is a continuous piecewise-linear model with a slope change at τ [**?**].

Mechanism term plus flexible correction (C_3). To allow a simple monotone mechanism shape with residual flexibility, we use a three-parameter saturation term plus a truncated-power correction with $K \in \{1, 2, 3\}$ fixed knots:

$$\hat{y}(t) = \beta_0 + \beta_1 \left(1 - e^{-\beta_2 t}\right) + \sum_{\ell=1}^{K} \gamma_\ell (t - \kappa_\ell)_+^3. \tag{7}$$

Knot locations κ_ℓ are fixed *a priori* (to keep a finite candidate family), so that K controls within-structure flexibility. C_1 (polynomial) includes degrees 0–2; C_2 (two-phase hinge / segmented regression) uses $\tau \in \{0.4, 0.5, 0.6\}$; and C_3 (mechanism + spline-like correction) uses 1–3 knots [**?**]. For each candidate, we fit by least squares and record the number of free parameters k_M.

Table 2
Candidate model family used in the benchmark (structure C and complexity f).

Model ID	Structure C	Complexity f	Free parameters k_M
C1_poly_deg0	single-phase polynomial	degree 0	1
C1_poly_deg1	single-phase polynomial	degree 1	2
C1_poly_deg2	single-phase polynomial	degree 2	3
C2_hinge_tau0.4	two-phase hinge	$\tau = 0.4$	3
C2_hinge_tau0.5	two-phase hinge	$\tau = 0.5$	3
C2_hinge_tau0.6	two-phase hinge	$\tau = 0.6$	3
C3_mech_spline_k1	mechanism + correction	1 knot	4
C3_mech_spline_k2	mechanism + correction	2 knots	5
C3_mech_spline_k3	mechanism + correction	3 knots	6

2.8. Robustness-Integrated Evidence and Choice (RIEC) score and protocol

Robustness-Integrated Evidence and Choice (RIEC) is a selection protocol designed to make *structure choice* explicit and auditable in repeated-measure settings. It combines three elements.

Robustness. We evaluate candidates under multiple decision-relevant CV regimes (forward-by-time, group-by-oil) and include random split as a negative control. Robustness is reported as the stability of the selected *structure C* across regimes.

Integrated evidence. For a given regime, we combine a parsimony term with out-of-sample evidence. We compute a BIC-style score using n_{eff}:

$$\text{BIC}(M) = -2\log L(\hat{\theta}_M) + k_M \log(n_{\text{eff}}), \tag{8}$$

We compute the Gaussian log-likelihood using the full observation count n (with $\hat{\sigma}^2 = \text{RSS}/n$ under least squares) and replace only the penalty term $\log(n_{\text{eff}})$ to reflect reduced information under within-oil dependence. and form the RIEC score

$$C_\lambda(M) = \text{BIC}(M) - \lambda(n_{\text{eff}})\log\big(\text{XPE}(M)\big), \tag{9}$$

which is equivalent to $\text{BIC}(M) + \lambda(n_{\text{eff}})\log(\text{XPE}(M)^{-1})$. Because $\log(\text{XPE}(M))$ increases with predictive improvement over baseline, the second term rewards out-of-sample evidence while the BIC term penalizes complexity.

We use a decaying weight $\lambda(n_{\text{eff}}) = c/\log(n_{\text{eff}})$ with $c = 2.0$ so that the influence of CV evidence decreases as information grows and the score approaches a pure information criterion in large samples.

Choice. We apply an explicit two-stage selection aligned with the structure/complexity decomposition:

$$f^*(C) = \arg\min_{f\in\mathcal{F}_C} C_\lambda(C, f), \qquad C^* = \arg\min_{C\in\mathcal{C}} C_\lambda\big(C, f^*(C)\big). \tag{10}$$

In the main analysis, C^* is chosen under forward-by-time evaluation (shelf-life extrapolation), and we report whether the same structure is selected under group-by-oil evaluation (robustness to decision context).

Algorithm 1 (RIEC protocol).

1. Specify a finite candidate set $\{M\} = \{(C, f)\}$ and choose one or more evaluation regimes r (forward-by-time; group-by-oil; random split as negative control).
2. For each regime r and each candidate M, compute $E_{\text{CV}}(M; r)$ and $\text{XPE}(M; r)$.
3. Estimate within-oil ICC ρ and compute n_{eff}.
4. Compute $\text{BIC}(M)$ using n_{eff} and then the RIEC score $C_\lambda(M; r)$ via Eq. **??**.
5. Perform two-stage choice under the primary regime r_0: pick $f^*(C)$ within each structure, then pick C^* across structures.
6. Report robustness: compare the selected structure across regimes and flag any disagreements.

2.9. Comparator criteria used in food-quality modelling

For context, we also report selections obtained from widely used model-selection baselines in food deterioration studies. These include the Akaike information criterion (AIC) [**?**], its small-sample correction (AICc) [**?**], and leave-one-unit-out cross-validation (LOOCV) as an extreme form of grouped CV. For a candidate M with k_M parameters

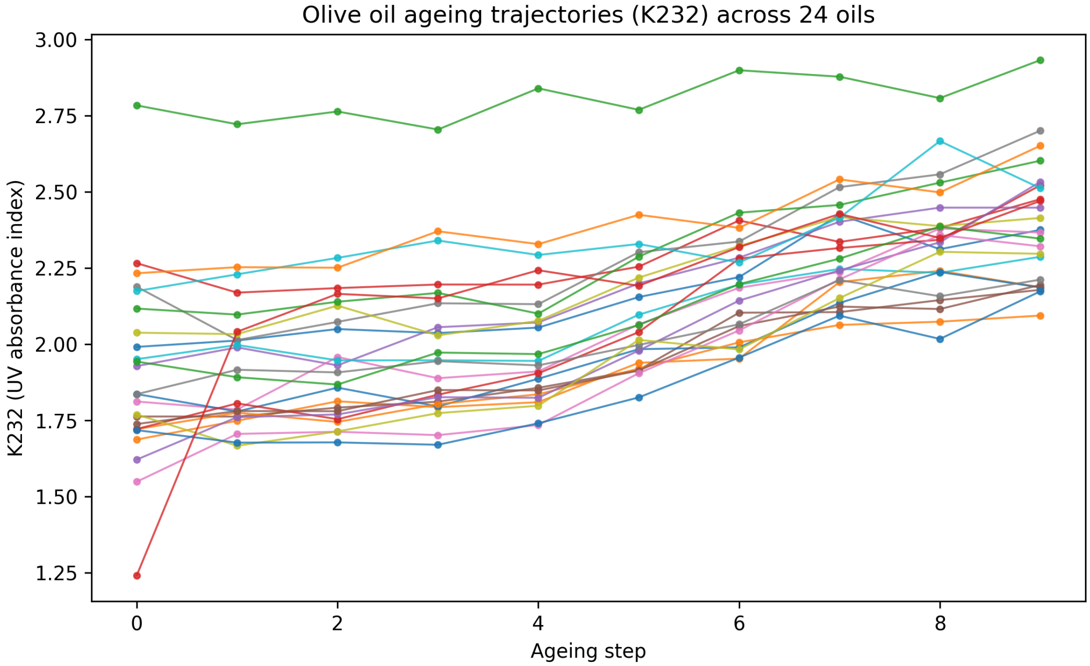

Figure 1: Ageing trajectories of K232 for 24 olive oils across 10 ageing steps. Each line is one oil.

and log-likelihood $\log L(\hat{\theta}_M)$, we define

$$\mathrm{AIC}(M) = -2\log L(\hat{\theta}_M) + 2k_M, \qquad \mathrm{AICc}(M) = \mathrm{AIC}(M) + \frac{2k_M(k_M+1)}{n^\star - k_M - 1}, \tag{11}$$

where $n^\star$ is the information size used for the correction. We report AICc both with $n^\star = n$ (naive) and with $n^\star = n_{\mathrm{eff}}$ as a pragmatic sensitivity analysis for repeated measures. Finally, we compute a leave-one-oil-out CV error $E_{\text{LOO-oil}}$ by holding out each oil trajectory in turn ($K = N$ folds). This grouped LOOCV avoids the row-level leakage that would occur if one left out individual time points within the same trajectory.

3. Results

3.1. Trajectories and effective sample size

Figure **??** shows within-oil trajectories for K232 across ageing steps. The measurement layer estimates within-oil correlation $\rho = 0.500$ and yields an effective sample size $n_{\mathrm{eff}} = 43.6$ (versus nominal $n = 240$), implying that treating points as independent would materially overstate evidence (Figure **??**).

3.2. Model selection differs by evaluation regime

Under forward-by-time evaluation (predict later ageing steps from earlier steps), RIEC selects a two-phase hinge model (C_2, $\tau = 0.4$) as the best trade-off between structure and complexity. This model achieves $E_{\mathrm{CV}} = 0.0496$ and XPE= 1.893, corresponding to a 47.2% error reduction versus the intercept baseline. Under group-by-oil evaluation (predict unseen oils), the same structure is selected, supporting robustness of the structural choice across decision contexts. Under random split, performance estimates become optimistic and structure ranking can differ, consistent with information leakage in trajectory data (see Supplementary Figure S3). Figure **??** summarizes the selection disagreement under forward-by-time evaluation.

3.3. RIEC stabilizes structure choice versus CV-only

In the group-by-oil regime, CV-only can prefer a more flexible candidate for a marginal 0.41% error gain, while RIEC selects the simpler structured hinge model. The difference is explained by n_{eff}-adjusted complexity penalties

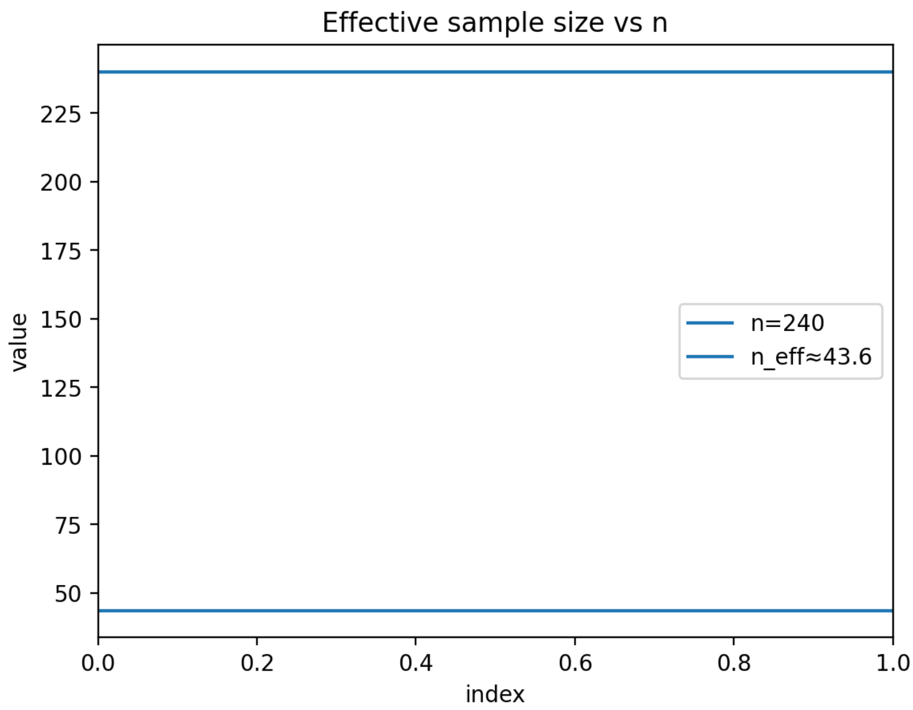

Figure 2: Nominal sample size n vs effective sample size n_{eff} under repeated measures (design-effect correction).

and the calibrated weight on out-of-sample evidence, yielding a more stable and interpretable structural conclusion under limited information.

3.4. Robustness diagnostics: stability across regimes and marginal gains

Table **??** summarizes the primary selection outcomes under each evaluation regime for all three UV-index targets (K232, K268 and ΔK). Under forward-by-time (forecasting), both CV-only and RIEC select the same two-phase hinge model ($\tau = 0.4$), which improves over the intercept baseline by XPE = 1.893 (47.2% error reduction). However, under group-by-oil evaluation (generalization to unseen oils), CV-only selects a more flexible mechanism-plus-correction candidate (C_3, $K = 1$) for a marginal 0.41% error gain relative to the hinge model, illustrating how a CV-only decision can drift toward additional flexibility for small improvements. By contrast, RIEC retains the hinge structure under both regimes because the n_{eff}-calibrated complexity penalty offsets such marginal gains, making the structural conclusion stable.

Rank stability across regimes is quantified in Table **??**. For this dataset, the RIEC ranking is identical across the three regimes, whereas CV-only rankings are highly regime dependent (e.g., Spearman $\rho_s = -0.37$ between forward-by-time and group-by-oil ranks). The near-zero correlation between group-by-oil and random-split CV ranks ($\rho \approx 0.02$) further motivates treating random splits as diagnostics rather than as decision-aligned estimates for trajectory data. Across all three targets (K232, K268 and ΔK), RIEC selects the same two-phase hinge model ($\tau = 0.4$) under all three regimes (Table **??**).

3.5. Benchmarking against AIC/AICc and leave-one-oil-out CV

To connect RIEC to practices commonly used in food-quality modelling, Table **??** contrasts the top choices from information criteria (BIC, AIC, AICc) and predictive-only criteria (CV-only and LOOCV). A key observation is that naive BIC computed with the nominal n is overly conservative in repeated measures and prefers a single-phase linear trend, whereas using n_{eff} restores preference for the stage-transition hinge structure. Predictive-only criteria can drift toward more flexible correction models when the error gains are extremely small (below 0.5%), which is practically important because such small average-error differences can still translate into noticeably different extrapolations near

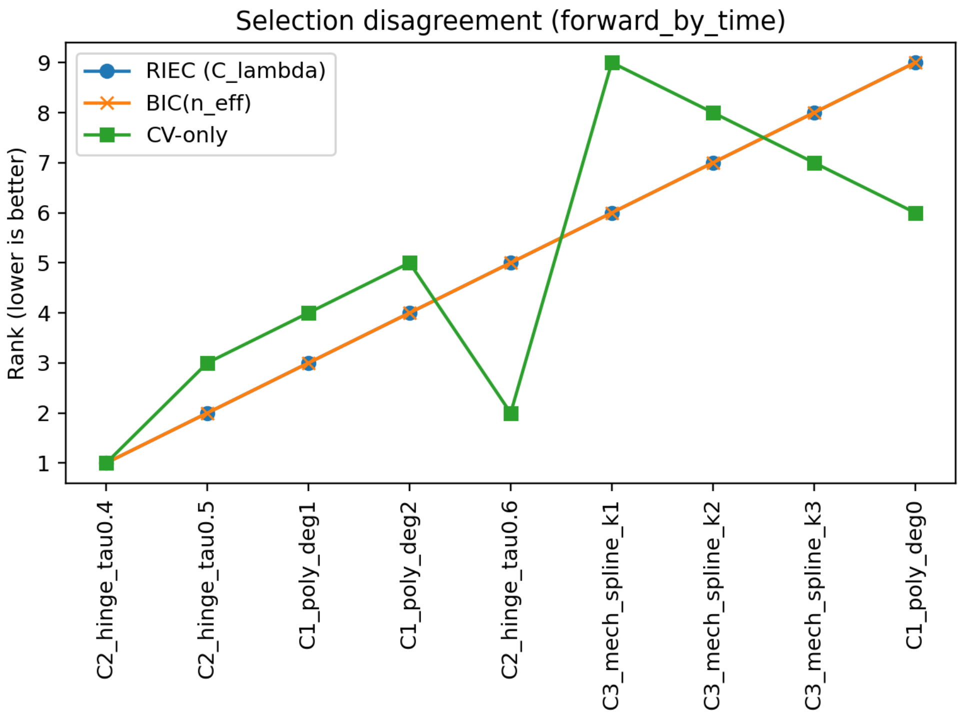

Figure 3: Model selection disagreement under forward-by-time evaluation: RIEC vs CV-only vs BIC(n_{eff}).

QC decision thresholds. RIEC makes this trade-off explicit by combining n_{eff}-calibrated parsimony with decision-aligned evidence and by reporting robustness across regimes.

4. Discussion

4.1. Why a two-phase structure is plausible for oxidation proxies

For edible oils, oxidation chemistry and measurement practice can lead to stage-like behavior (e.g., an initial lag followed by faster change) [**?**]. In UV spectroscopy of olive oil, K232 and K268 are linked to conjugated diene and triene systems, and ΔK captures local spectral curvature; these indices are routinely used as quality proxies [**?**]. The hinge structure selected by RIEC captures stage transitions without resorting to high-degree polynomials that are hard to interpret and may extrapolate poorly.

4.2. Why evaluation design matters

Random splits can leak within-oil trajectory information into both train and test sets, producing overly optimistic errors and potentially misleading structure choices. Forward-by-time evaluation aligns with shelf-life and forecasting decisions, while group-by-oil aligns with generalization to new lots or suppliers. These concerns mirror established cautions about using standard CV blindly on dependent or ordered data [**?**].

4.3. What RIEC adds beyond standard practices

BIC provides an explicit complexity penalty but can be miscalibrated when the effective information is much smaller than n due to repeated measures [**? ?**]. CV directly estimates out-of-sample performance but can be high-variance in small samples and may prefer extra flexibility for tiny error differences [**? ?**]. RIEC operationalizes a practical bridge: it adjusts information content via n_{eff}, and it uses a decaying weight so that predictive evidence has more influence when information is scarce but fades as information grows.

Table 3
Selections and performance across evaluation regimes for each target UV index (K232, K268 and ΔK). Effective sample sizes used in RIEC were target-specific (K232: $n_{\text{eff}} = 43.6$; K268: $n_{\text{eff}} = 105.7$; ΔK: $n_{\text{eff}} = 167.6$).

Panel A: Baseline and RIEC.

Regime	Baseline E_{CV}	RIEC choice	E_{CV}	XPE
K232				
Forward-by-time ($K = 4$)	0.0939	C_2 hinge ($\tau = 0.4$)	0.0496	1.893
Group-by-oil ($K = 5$)	0.0817	C_2 hinge ($\tau = 0.4$)	0.0521	1.568
Random split ($K = 5$)	0.0811	C_2 hinge ($\tau = 0.4$)	0.0525	1.544
K268				
Forward-by-time ($K = 4$)	0.0081	C_2 hinge ($\tau = 0.4$)	0.0024	3.316
Group-by-oil ($K = 5$)	0.0057	C_2 hinge ($\tau = 0.4$)	0.0018	3.128
Random split ($K = 5$)	0.0056	C_2 hinge ($\tau = 0.4$)	0.0017	3.264
ΔK				
Forward-by-time ($K = 4$)	5.52×10^{-5}	C_2 hinge ($\tau = 0.4$)	1.12×10^{-5}	4.927
Group-by-oil ($K = 5$)	3.66×10^{-5}	C_2 hinge ($\tau = 0.4$)	7.76×10^{-6}	4.710
Random split ($K = 5$)	3.65×10^{-5}	C_2 hinge ($\tau = 0.4$)	7.79×10^{-6}	4.691

Panel B: CV-only (predictive-only selection).

Regime	CV-only choice	E_{CV}
K232		
Forward-by-time ($K = 4$)	C_2 hinge ($\tau = 0.4$)	0.0496
Group-by-oil ($K = 5$)	C_3 mech+spline ($K = 1$)	0.0519
Random split ($K = 5$)	C_2 hinge ($\tau = 0.4$)	0.0525
K268		
Forward-by-time ($K = 4$)	C_2 hinge ($\tau = 0.4$)	0.0024
Group-by-oil ($K = 5$)	C_3 mech+spline ($K = 3$)	0.0018
Random split ($K = 5$)	C_2 hinge ($\tau = 0.4$)	0.0017
ΔK		
Forward-by-time ($K = 4$)	C_2 hinge ($\tau = 0.4$)	1.12×10^{-5}
Group-by-oil ($K = 5$)	C_3 mech+spline ($K = 3$)	7.62×10^{-6}
Random split ($K = 5$)	C_2 hinge ($\tau = 0.4$)	7.79×10^{-6}

Table 4
Rank stability across regimes (Spearman correlation of candidate ranks). For CV-only, ranks are based on E_{CV}; for BIC(n_{eff}) on the information criterion; and for RIEC on C_λ.

Method	Forward vs group	Forward vs random	Group vs random
CV-only	-0.37	0.73	0.02
BIC(n_{eff})	1.00	1.00	1.00
RIEC	1.00	1.00	1.00

4.4. Food-industry value: QC-relevant model choice and auditability

In many food applications, the end user of a deterioration model is a QC scientist or process engineer who must justify shelf-life labels and compliance decisions. Even when average prediction errors differ only marginally, extrapolated trajectories can cross acceptance thresholds at noticeably different times, especially near a phase transition. For olive oil, UV indices are part of routine compliance screening [**?**]; in this setting, selecting an over-flexible curve can "chase" within-trajectory noise and shift the implied time-to-limit. RIEC is designed to reduce this risk pragmatically: it prefers interpretable structures unless there is clear out-of-sample evidence for extra flexibility, and it requires that conclusions be reported under decision-relevant evaluation regimes.

Table 5
Comparison with common selection baselines for K232. BIC(n) uses the nominal sample size, while BIC(n_{eff}) uses the effective sample size from Eq. (**??**). AIC and AICc are computed from the Gaussian likelihood of each candidate; AICc is reported with $n^\star = n$ and with $n^\star = n_{\text{eff}}$. LOOCV is implemented as leave-one-oil-out grouped CV.

Criterion	Selected model	Metric / comment
RIEC (primary: forward-by-time)	C2 hinge ($\tau = 0.4$)	$E_{\text{CV}} = 0.0496$; structure remains selected under group-by-oil.
BIC(n) (naive)	C1 polynomial (deg 1)	Overly conservative under repeated measures; forward-by-time $E_{\text{CV}} = 0.0503$ (1.44% worse than hinge).
BIC(n_{eff})	C2 hinge ($\tau = 0.4$)	Dependence-corrected complexity restores the stage-transition structure.
AIC / AICc(n) / AICc(n_{eff})	C2 hinge ($\tau = 0.4$)	Same top choice here, but these criteria do not encode decision-aligned CV regimes or leakage diagnostics.
CV-only (group-by-oil, $K = 5$)	C3 mech+spline ($K = 1$)	$E_{\text{CV}} = 0.0519$, a marginal 0.41% gain vs hinge, illustrating drift to extra flexibility for tiny gains.
LOOCV (leave-one-oil-out, $K = N$)	C3 mech+spline ($K = 3$)	$E_{\text{LOO-oil}} = 0.0552$, a marginal 0.28% gain vs hinge; differences are tiny but the selected model is more complex.

4.5. Practical guidance for applying RIEC in food-quality studies

While RIEC is presented here as a benchmark protocol, it can be used as a pragmatic "decision wrapper" around standard candidate modelling. In practice, we recommend the following workflow for repeated-measure deterioration datasets:

- **Define the decision context first.** If the goal is shelf-life extrapolation, use a forward-by-time regime as the primary target; if the goal is supplier/lot transfer, use group-by-unit as the primary target.
- **Keep the candidate set finite and interpretable.** Include a small number of mechanistically plausible structures (e.g., one-phase vs. two-phase) and a limited set of within-structure flexibilities so that selection outcomes are auditable rather than purely data-driven.
- **Report dependence and effective information.** Estimate ρ and n_{eff} and explicitly state whether complexity penalties are calibrated to n or to n_{eff}.
- **Use random splits only diagnostically.** If random-split errors are much lower than blocked or grouped errors, or if rankings change drastically, this is evidence that naive CV may be misaligned with the intended decision.

These steps aim to prevent overconfident conclusions from trajectory leakage and to make the rationale for model choice transparent to reviewers and practitioners.

4.6. Limitations and next steps

This benchmark uses UV proxies rather than full chemical panels (e.g., peroxide value, acid value, volatile markers, or sensory scores), and ageing is accelerated at 60 °C [**? ?**]. While we analyse three UV indices (K232, K268 and ΔK), all originate from the same measurement modality; direct translation to ambient shelf-life would require additional datasets and, ideally, multi-temperature models (e.g., Arrhenius-type extensions). The RIEC protocol itself is general: it applies to any repeated-measure deterioration dataset once the dependence "unit" (cluster) and the decision context (forecasting along time vs. generalization to new units) are specified. We used a Kish design-effect correction for a transparent n_{eff}; future work should compare this calibration to likelihood-based mixed-effects information criteria and to resampling-based uncertainty estimates on richer and more unbalanced longitudinal datasets. Finally, broader empirical validation on other food systems (other edible oils, dairy, meat, plant-based products) and other indicator types would strengthen evidence for general applicability and help establish community reporting norms.

5. Conclusions

In a public olive-oil UV ageing dataset with repeated measures, treating rows as independent can inflate evidence and destabilize model choice. RIEC provides a mechanism-aware, audit-ready model selection workflow: it separates structure from complexity, corrects information content via an effective sample size, and combines parsimony with out-of-sample evidence. Across decision-aligned evaluation regimes, RIEC identifies a simple two-phase hinge structure that yields strong forecasting performance while remaining interpretable for food-quality practice.

Data and code availability

Data: The study uses a public dataset available on Mendeley Data (DOI: 10.17632/g6y69g8gwm.1) [**?**]. Code: A runnable reproduction package is available at `https://github.com/lijincheng494gmail/crfs-reproduce` (archive: `https://github.com/lijincheng494gmail/crfs-reproduce/raw/main/riec_crfs_reproduce.zip`). A brief runbook with one-click commands and expected outputs is provided in the submission files (see the Reproducibility links & instructions document).

CRediT authorship contribution statement

Jincheng Li: Modelling experiments, Methodology, Software, Formal analysis, Writing—original draft, Writing—review & editing. Xifeng Li and Jinlong Ji: Domain guidance and calibration (food and edible oils), Validation (domain applicability check), Writing—review & editing.

Declaration of competing interest

The authors declare no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary File 3

Domain Background, Reader Guide, and How to Connect RIEC to Food-QC Decisions

(for the manuscript "Robustness-Integrated Evidence and Choice (RIEC)")

1 Who this file is for

This file is written for **domain readers** (food science, fats and oils, industry QC) who want to understand:

- what the UV indices (K232, K268, ΔK) represent at a practical level,
- why repeated-measure trajectories change how model evaluation should be done,
- and how to interpret Robustness-Integrated Evidence and Choice (RIEC) outputs as *actionable* information (not just "better statistics").

It also serves as a "bridge" for method reviewers by explicitly mapping domain decision questions to the evaluation regimes used in the manuscript.

2 Practical meaning of the UV indices (what they are used for)

2.1 K232 and K268 in plain language

In olive oil quality control, UV absorbance indices are widely used as rapid oxidation proxies.

- **K232** is typically associated with early oxidation products (often discussed as conjugated diene-related absorbance).
- **K268** is typically associated with later-stage oxidation or more advanced oxidation-related absorbance changes.

The key point for this paper is *not* a new chemical mechanism, but that these indices provide standardized, low-cost signals that change over time and are used to monitor quality and compliance.

2.2 ΔK as an auxiliary UV metric

ΔK (reported as `DeltaK` in the dataset) is commonly used as an auxiliary diagnostic index in UV-based assessment workflows. In practice, multiple UV indices are often monitored simultaneously to support decisions.

3 Why trajectories and repeated measures matter in shelf-life / QC modelling

3.1 Repeated measures create dependence

In this benchmark, each oil is measured repeatedly across ageing steps, forming a within-oil trajectory. Measurements from the *same* oil are correlated (shared raw material, processing history, matrix effects, etc.). Treating each row as independent can lead to overconfident model choice.

3.2 The leakage problem (why random splits can be misleading)

If we randomly split rows into training and test sets, both sets may contain points from the *same* oil trajectory. A flexible model can then "recognize" an oil-specific baseline level or curvature from training points and appear unrealistically accurate on test points from the same oil. This is why the manuscript includes random splits only as a **negative control**.

4 Decision questions → evaluation regimes (the core linkage)

Different stakeholders ask different questions. RIEC makes these explicit by reporting multiple evaluation regimes. Table 1 is a suggested mapping.

Table 1: **Table S6.** Mapping of decision questions to evaluation regimes and typical outputs to report.

Decision question (domain)	Regime to prioritize	What output is most actionable
"Can early measurements predict later ageing?" (shelf-life extrapolation / forecasting)	Forward-by-time	Forecast accuracy on later steps; stage-transition location τ if a two-phase structure is selected; uncertainty / caution flags when n_{eff} is small.
"Will the model generalize to a new lot/supplier?" (unseen oils)	Group-by-oil	Errors on held-out oils; stability of selected structure across oils; whether structure choice changes compared with forward-by-time.
"Quick benchmark only" (not a real decision)	Random split (diagnostic only)	Use only to detect leakage: if random-split error is far smaller than group-aware errors, the benchmark is likely optimistic.

5 How to interpret the three candidate structures as "worldviews"

RIEC separates **structural choice** from **within-structure flexibility**.

5.1 Single-phase polynomial (C_1): smooth trend

C_1 assumes the trajectory can be captured by a single smooth trend across the entire ageing window. It is easy to communicate, but may miss genuine stage changes.

5.2 Two-phase hinge (C_2): explicit stage transition

C_2 assumes there is a change in slope at a transition point $\tau \in [0, 1]$. In QC language, this corresponds to "early stable" → "later faster change" behavior. If C_2 is selected robustly, it provides a **simple and interpretable** stage-transition summary.

5.3 Mechanism-inspired saturation + spline correction (C_3): structured + flexible

C_3 contains a mechanistic saturating component plus a smooth correction term. This can capture a broader range of shapes, but also increases risk of overfitting when effective information is limited. RIEC is designed to prevent small error gains from automatically justifying large flexibility increases.

6 Interpreting RIEC outputs in domain terms

6.1 What does XPE tell me?

XPE compares a candidate against a baseline intercept model:

- XPE = 1 means "no improvement over baseline".
- Larger XPE means larger predictive improvement.

Because it is a ratio, it is easier to compare across endpoints (K232 vs K268 vs ΔK) than raw MSE.

6.2 What does n_{eff} tell me?

n_{eff} is a compact way to communicate how much *independent information* is available after accounting for within-oil dependence. A small n_{eff} suggests that "240 rows" do not behave like 240 independent experiments; therefore, high-flexibility models should be penalized more strongly.

6.3 What does "robust" mean here?

A practically robust conclusion is one where **structural choice** (e.g., "two-phase is needed") does not change dramatically across decision-relevant regimes. If a conclusion depends on a random split but disappears under group-by-oil or forward-by-time, it is less trustworthy for real QC use.

7 How to report results so food/QC readers can use them

A minimal domain-facing report for each endpoint can include:

- The selected structure (C_1 vs C_2 vs C_3) under the primary regime (usually forward-by-time).
- If C_2 is selected: the estimated transition point τ and a short interpretation ("transition around 40% of the ageing window").
- XPE and E_{CV} under the primary regime.
- A robustness statement: whether the selected structure matches under group-by-oil evaluation.

8 Limitations and how to generalize to other foods/endpoints

8.1 Accelerated ageing vs real-time shelf life

This benchmark uses accelerated ageing (60°C). While accelerated protocols are valuable for screening, a direct mapping from accelerated time to real-time shelf life typically requires additional assumptions (e.g., Arrhenius-type temperature dependence) and/or multi-temperature experiments.

8.2 Other food systems and endpoints

The protocol can be adapted to:

- other oils/fats (different cultivars, refining levels, storage conditions),
- other endpoints (peroxide value, anisidine value, free fatty acids, sensory scores, volatile markers),
- other products with repeated measurements (dairy oxidation, meat color, texture degradation, etc.).

The adaptation recipe is:

1. define the repeated-measure unit (batch/lot/product unit),
2. pick the evaluation regime that matches the intended decision,
3. specify a finite candidate family consistent with domain knowledge,
4. quantify dependence (estimate ρ) and compute n_{eff} for penalty calibration,
5. report robustness by comparing at least two decision-relevant regimes.

9 Quick reading roadmap

If you are a domain reader (10–15 min): focus on the main manuscript figures (trajectories, n vs n_{eff}, and selection under forward-by-time) plus the short "How to report results" section above.

If you are a methods reader (20–30 min): read Supplementary File 2 (definitions), then Supplementary File 1 (exact folds and artifacts), then inspect the exported selection tables.

Supplementary File 1

Experiment Design, Implementation Details, and Reproducibility

(for the manuscript "Robustness-Integrated Evidence and Choice (RIEC)")

1 Purpose and scope

This supplementary file provides an audit-ready description of the computational pipeline used in the Robustness-Integrated Evidence and Choice (RIEC) benchmark study. It expands on the *Evaluation regimes* and *Implementation details* sections of the main manuscript, and is written for readers who want to (i) understand exactly what each evaluation regime tests, and/or (ii) reproduce figures and selection tables from code.

2 Data processing and schema

2.1 Data unit and repeated-measure structure

The benchmark uses an open olive-oil accelerated ageing dataset with **24 oils** measured at **10 ageing steps** (accelerated ageing at 60°C), for a total of $n = 240$ **rows**. The repeated-measure unit is the **oil** (cluster) identified by `oil_id`. Each oil contributes a within-oil trajectory across ageing steps.

2.2 Processed long-format table

All model selection runs use a processed long-format CSV (`venturini_uv_long.csv` in the reproducibility artifacts). The core fields used by the protocol are summarized in Table 1.

Table 1: **Table S1.** Processed long-format data schema used for model selection runs.

Field	Description
oil_id	Oil identifier (cluster unit). Each oil has 10 repeated measurements.
ageing_step	Integer step index (0–9) under accelerated ageing.
t	Normalized time used in modelling: $t = \text{ageing\_step}/9 \in [0, 1]$.
K232	UV absorbance index at 232 nm (oxidation proxy).
K268	UV absorbance index at 268 nm (oxidation proxy).
ΔK	Derived UV index reported in the dataset (DeltaK column).

2.3 Normalized time used in modelling

We model each target as a function of a normalized time variable

$$t = \frac{\text{ageing\_step}}{9} \in \left\{0, \tfrac{1}{9}, \ldots, 1\right\}.$$

This ensures that candidate models (e.g., hinge locations τ) live on a common, unit-free scale.

3 Candidate family and baseline

3.1 Finite candidate set

We compare a finite candidate family written as $M = (C, f)$, where C is a *structural form* and f indexes *within-structure complexity*. The benchmark includes nine candidates (Table 2), matching Table 1 of the main manuscript.

Table 2: **Table S2.** Candidate model family used in the benchmark (finite set). Here $M = (C, f)$ denotes a candidate with structural form C and within-structure complexity f; k_M is the number of free parameters.

Model ID	Structure C	Complexity f	k_M	Notes
C1_poly_deg0	single_phase_poly	deg0	1	degree=0
C1_poly_deg1	single_phase_poly	deg1	2	degree=1
C1_poly_deg2	single_phase_poly	deg2	3	degree=2
C2_hinge_tau0.4	two_phase_hinge	tau0.4	3	hinge_at_frac=0.4
C2_hinge_tau0.5	two_phase_hinge	tau0.5	3	hinge_at_frac=0.5
C2_hinge_tau0.6	two_phase_hinge	tau0.6	3	hinge_at_frac=0.6
C3_mech_spline_k1	mech_plus_spline	k1	4	knots=[0.5]
C3_mech_spline_k2	mech_plus_spline	k2	5	knots=[0.33, 0.66]
C3_mech_spline_k3	mech_plus_spline	k3	6	knots=[0.25, 0.5, 0.75]

3.2 Baseline definition

The baseline model is `C1_poly_deg0` (an intercept-only model). Baseline-normalized improvements are reported via the relative improvement ratio (XPE; defined in the main manuscript and Supplementary File 2).

4 Evaluation regimes and cross-validation splits

4.1 Why multiple regimes?

Different food-science decisions correspond to different generalization questions:

- **Shelf-life extrapolation / forecasting**: "Given early ageing steps, how well can we predict later steps?" → forward-by-time.
- **Generalization to new lots/suppliers**: "Given some oils, how well do we predict a new oil trajectory?" → group-by-oil.
- **Negative control**: random row-level splits can *leak* within-oil trajectory information and yield optimistic errors; we include random splits only to diagnose such leakage.

4.2 Forward-by-time (primary in the manuscript)

Forward-by-time uses an expanding-window blocked design with $K = 4$ folds (Table 3). All folds respect time order: training uses earlier steps and testing uses later steps.

4.3 Group-by-oil (robustness check)

Group-by-oil uses $K = 5$ folds that hold out entire oils. In each fold, the test set contains all 10 ageing steps for the held-out oils. The exact held-out oil IDs are shown in Table 4.

Table 3: **Table S3.** Forward-by-time cross-validation design (expanding-window). Training steps always precede test steps, mimicking forecast-style shelf-life extrapolation.

Fold	Train ageing steps	Test ageing steps
1	0–1	2–3
2	0–3	4–5
3	0–5	6–7
4	0–7	8–9

Table 4: **Table S4.** Group-by-oil cross-validation design. Each fold holds out entire oils (all ageing steps for those oils), representing generalization to unseen lots/suppliers.

Fold	Held-out oil IDs
1	J, P, Q, S, T
2	G, H, K, R, X
3	A, D, F, M, U
4	C, E, L, O, W
5	B, I, N, V

4.4 Random split (diagnostic negative control)

Random split uses $K = 5$ folds and ignores grouping/time order (row-level K-fold). In our configuration files, we set `shuffle=true` and a fixed `random_state=123` to make this diagnostic repeatable. **Important:** random split is *not* used to support any substantive conclusions; it is a sanity check for leakage.

5 Effective sample size and dependence manifest

To calibrate complexity penalties under repeated measures, we estimate within-oil dependence via an intraclass correlation (ICC) and compute an effective sample size n_{eff} (details and derivation are in Supplementary File 2). The estimated values (computed from the K232 target) are shown in Table 5.

Table 5: **Table S5.** Estimated dependence and effective information (computed from the K232 target as described in the manuscript).

Quantity	Value
n (rows)	240
$\bar{m}$ (avg. repeats / oil)	10.0
ρ (ICC)	0.500
Design effect deff	5.503
Effective sample size n_{eff}	43.6

6 Model fitting and numerical details

6.1 Loss function

All evaluation regimes use mean squared prediction error (MSE) as the fold-level loss. The cross-validated error $E_{\text{CV}}(M)$ is computed as the average of fold-level MSEs (see Supplementary File 2 for the full definition).

6.2 Fitting procedures

- Candidates in structures C_1 (single-phase polynomial) and C_2 (two-phase hinge) are linear-in-parameters and are fitted by ordinary least squares.
- Candidates in C_3 (mechanism-inspired saturation plus spline correction) are fitted by nonlinear least squares. We impose a nonnegativity constraint on the saturation rate parameter to avoid nonphysical "growth" behavior in the saturating component.

7 Audit artifacts produced by the reproducible run

A reproducible run exports:

- **Split manifests:** JSON files listing train/test row indices for each fold and each regime (e.g., `cv_splits_forward_by_time.json`).
- **Measurement manifest:** dependence estimates and n_{eff} (`measurement_manifest.json`).
- **Selection tables:** candidate-level summaries for each regime including E_{CV}, XPE, BIC(n), BIC(n_{eff}), and the RIEC score C_λ (e.g., `selection_table_forward_by_time.csv`).
- **Configuration snapshots:** YAML files describing the candidate set and split settings (`configs/`).

These artifacts are intended to make the reported results fully auditable even without rerunning the full pipeline.

8 One-click reproduction (external code package)

The submission bundle contains the audit artifacts above and a minimal run guide. The full runnable code is hosted externally (e.g., on GitHub) and can be executed with a single shell command after installing dependencies.

8.1 Recommended repository link format

We recommend pointing reviewers to a stable repository URL plus a tagged release/commit hash. For example:

- GitHub user page: https://github.com/lijincheng494gmail
- Repository (example): https://github.com/lijincheng494gmail/<REPO_NAME>
- Release/tag: <RELEASE_TAG> (or commit <COMMIT_SHA>)

8.2 One-click run command

A minimal one-click reproduction procedure is:

```
cd <PROJECT_ROOT>/experiment/RIEC-JAFC-Olive
python3 -m venv .venv
source .venv/bin/activate
python -m pip install --upgrade pip
python -m pip install -r requirements.txt
bash run_mvp.sh
```

8.3 Sanity checks (expected outputs)

After a successful run, the following sanity checks should hold:

- The processed table contains 240 rows (24 oils $\times$ 10 ageing steps).

- Under forward-by-time evaluation for the K232 target, the top RIEC model is `C2_hinge_tau0.4` (two-phase hinge with $\tau = 0.4$).
- Output folders include regenerated figures/tables and the audit artifacts (split and measurement manifests).

9 How to adapt this protocol to another dataset

To apply the same reproducible protocol to another repeated-measure deterioration dataset, ensure that:

1. A **cluster identifier** is defined (e.g., product unit, batch, lot).
2. A **decision-aligned evaluation regime** is specified (forecasting vs generalization).
3. A **finite candidate family** is written down *before* looking at test results.
4. Dependence is quantified (e.g., ICC) and an effective information size n_{eff} is computed for complexity calibration.

Declaration of Generative AI and AI-assisted Technologies in the Writing Process

The authors declare that generative AI tools were used during the preparation of this manuscript **only** to assist with English language polishing and to check the consistency of the final submission files (e.g., completeness and naming). All scientific content, data analysis, interpretations, and conclusions are those of the authors.

The authors reviewed and edited all text as needed and take full responsibility for the content of the submitted work.

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